

Quantifying the Sensitivity of Quantitative Trade Models*

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Abstract

Modern quantitative trade models emphasize cross-country heterogeneity in productivity across goods as the source of trade and its benefits. In workhorse versions, this heterogeneity is assumed to follow specific functional forms, such as i.i.d. Frechet or Pareto. We provide a systematic exploration of how sensitive the gains from trade estimated from these models are to relaxing this assumption—first, with a new formula for sharp bounds on gains from trade across all possible distributions, and then with a numerical characterization for the case where distributions must be close to the workhorse form. Importantly, for each candidate distribution we ensure that its implied trade model is consistent with standard estimation methods such as the ability to match all bilateral trade data points and an exogeneity assumption. Our estimated bounds are wide, even for small departures from the canonical model: the maximal gain is about four times larger and the minimal gain is zero. Imposing high-level restrictions on the distributions considered—for example, maintaining Frechet or Pareto marginals while relaxing independence, or vice versa—does little to shrink these bounds.

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1 Introduction

Major advances in the field of international economics have come recently from the development of quantitative trade models: general equilibrium models that are designed to match empirical patterns about factual scenarios and make predictions about economic responses to counterfactual ones.¹ A core ingredient is the extent to which productive capabilities in the model are heterogeneous across goods and countries. Indeed, the exploitation of this heterogeneity—where countries import the goods that cannot be made cheaply at home—is what gives rise to trade and its benefits in the first place. However, quantifying this heterogeneity poses a foundational challenge: how can one hope to measure a country’s ability to produce the goods it doesn’t produce?

To make progress, researchers have posited specific functional forms to describe production heterogeneity around the world. Two choices that result in so-called gravity models—the independent and identically distributed (i.i.d.) Frechet distribution used by [Eaton and Kortum \(2002\)](#) for competitive models and the i.i.d. Pareto distribution used by [Chaney \(2008\)](#) for monopolistically competitive models—have become enormously influential for workhorse quantitative modeling. But how would economists’ predictions about counterfactuals, such as the size of the gains from trade, change if alternative forms of production heterogeneity were considered instead? The goal of this paper is to conduct a comprehensive quantification of the sensitivity of estimated gains from trade in quantitative trade models as we allow for departures from these workhorse assumptions.

Our approach is disciplined by three important constraints. First, in order to focus on the role of production heterogeneity, we hold constant other model components, such as preferences and factor market structure. Second, in line with standard practice for quantitative modeling, we ensure that all bilateral trade flow data points are matched exactly as we vary countries’ productivity distributions—ensuring that empirical facts such as the extent of trade openness, the fit of the gravity equation, and the correlation between trade flows and tariffs are always replicated too. Finally, we impose a standard identification assumption that observed trade costs are orthogonal to unobserved ones.

Then, subject to these constraints, we derive new formulae for the maximal and minimal bounds on countries’ gains from trade that are possible across all underlying productivity distributions. We also compute similar bounds for the case where admissible distributions must be “similar” to (i.e. have low statistical divergence from) the gravity

¹Two recent chapters in the *Handbook of International Economics*, [Costinot and Rodríguez-Clare \(2014\)](#) and [Caliendo and Parro \(2022\)](#), provide surveys of this practice. Analogous progress on quantitative spatial models of intra-national trade is also surveyed in two recent chapters of the *Handbook of Regional and Urban Economics*, [Allen and Arkolakis \(2025\)](#) and [Redding \(2025\)](#).

model versions. Our main finding is that the estimated gains from trade in quantitative models are highly sensitive to alternative distributions of production heterogeneity.

Beginning with the distribution-agnostic case, our new analytical result states that (provided a mild data condition holds, as it always does in our sample) any country's gains from trade are bounded in a very simple manner. The maximal gain is given by a simple formula involving one data input (the country's openness) and one preference parameter (the elasticity of substitution across goods). For France, a representative case, this upper bound is 4.4 times larger than the typical gravity model benchmark. There are therefore data-consistent versions of underlying productivity distributions for which gains from trade can be very large. On the other hand, our lower bound is zero, so it is also possible that there are no gains from trade at all.

While these wide bounds allow for distributions that may be arbitrarily different from those of the gravity model workhorses of [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) our next result suggests that similarly wide bounds are possible even for distributions that are restricted to be "close" to these benchmarks. Our approach here draws on recent advances in distribution-robust econometrics, beginning with [Hansen and Sargent \(2001\)](#). This allows us to quantify bounds on the gains from trade when distributions must lie within a fixed level of divergence (a measure of the similarity of any two probability distributions) from the gravity case. Focusing on divergence levels that are analogous to other robustness checks that are routinely performed on workhorse models, we find that France's gains from trade can range between zero and 3.9 times the gravity model value. So they can essentially span most of the range (i.e. zero to 4.4 times) that exists when all distributions are allowed for.

It is of course infeasible to evaluate every one of the infinite number of distributions within any given neighborhood that is close to the gravity model. And doing so is harder still when every candidate distribution must be tested for whether it can generate a world-wide equilibrium whose trade flows match all data points and that it is consistent with standard gravity model estimation requirements. To circumvent this challenge we deploy the approach in [Christensen and Connault \(2023\)](#), who consider an optimization problem over all distributions of a random variable within a given divergence neighborhood, where the objective and constraints are expectations of (known functions of) the random variable. Following this logic, we write our problem—maximize or minimize gains from trade subject to the constraints of model equilibrium, matching all trade data, and an orthogonality assumption—as one with an inner loop that searches over distributions (given all unknown model parameters and market-clearing equilibrium variables in the factual and counterfactual equilibria) and an outer loop that searches over the pa-

rameters and equilibrium variables. The inner loop is infinite-dimensional but benefits from a dual formulation proposed by [Christensen and Connault \(2023\)](#) that is both finite-dimensional and convex. The outer loop is high-dimensional and typically non-convex but benefits from a dimensionality-reduction procedure that we develop. Putting these results together, the program becomes feasible.

Given that we find so little robustness of quantitative models, even for small amounts of departure from the gravity model, it is natural to restrict the admissible sets of productivity distributions further. Our final set of results does this by imposing high-level requirements that are guided by three core features of the benchmark i.i.d. distributions used in gravity models: (i) countries take draws independently from one another; (ii) countries all have a common marginal distribution; and (iii) the particular marginal distribution that is common is the Frechet form in [Eaton and Kortum \(2002\)](#) and the Pareto form in [Chaney \(2008\)](#). We show how to extend our baseline analysis—varying the model’s production heterogeneity while matching all trade flows and standard orthogonality conditions—while also imposing constraints (i) through (iii), one at a time or in combination, in order to learn which particular features of the benchmark distribution do the most to constrain the model’s predictions.

Across a wide range of restrictions imposed, the broad message that emerges is that the bounds on gains from trade remain wide despite requiring that productivity distributions satisfy reasonable high-level restrictions. We begin by establishing that for the case of monopolistically competitive models these restrictions have no direct bite at all because competition is never head-to-head on the same good. Turning to the competitive case, where such head-to-head competition is central, we first extend our previous distribution-agnostic upper bound formula to cases where the productivity distribution must satisfy requirements such as independence, common (and Frechet) marginals, or even that countries draw i.i.d from some common marginal distribution. Each case requires a different data condition to hold, but these conditions are verified in our sample.

We then move on to evaluate neighborhoods that are close to the gravity model. In this setting, the bounds on gains from trade at fixed neighborhood size do shrink in the face of high-level restrictions, but the effects are modest. For example, when we impose conditions that approximate full independence our upper bound shrinks from 3.9 times the baseline to 3.3 times the baseline (i.e. by 18%). And this is similar for the case where we impose zero correlation (18% shrinkage), common but flexible marginals (11%), common and [Eaton and Kortum \(2002\)](#) marginals (11%), and the i.i.d. condition of both independence and common but flexible marginals (24%). Perhaps surprisingly, this set of results implies that one can depart from Ricardian models with i.i.d. Frechet marginals in either

of two obvious directions—dropping independence and keeping the identical Frechet marginal, or keeping independence and dropping Frechet marginals—and achieve similarly modest reductions in sensitivity in either case.

Related literature. Our analysis builds on previous explorations of how quantitative trade models behave under varying assumptions about production heterogeneity.

In particular, our approach is heavily influenced by that of [Arkolakis et al. \(2012\)](#), who proposed that the sensitivity of conclusions obtained across models should be conducted while conditioning on two requirements: (i) that the models are calibrated to match the same underlying trade data; and (ii) that the models are estimated from a gravity equation regression that assumes the same underlying orthogonality condition between observed and unobserved trade costs.² Indeed, those authors show that, once these two conditions hold, counterfactuals are strikingly robust across the four types of production heterogeneity they study—the Frechet distribution of [Eaton and Kortum \(2002\)](#) and the degenerate distribution of [Armington \(1969\)](#) in the competitive case, and the Pareto distribution of [Chaney \(2008\)](#) and the degenerate distribution of [Krugman \(1980\)](#) in the monopolistically competitive case. We maintain these same two requirements but then broaden the set of productivity distributions considered to include all such distributions in order to fully assess the robustness uncovered by [Arkolakis et al. \(2012\)](#).

Our assessment is therefore complementary to other work that has evaluated productivity distributions that depart from the four members of the [Arkolakis et al. \(2012\)](#) class. On the competitive side, for example, [Lind and Ramondo \(2023\)](#) emphasize productivity distributions in which countries’ draws exhibit rich forms of dependence. We embrace this direction by considering arbitrary distributions, as well as a restricted class that allows for any form of dependency but where countries’ marginal distributions are the same as in the workhorse model. [Adao et al. \(2017\)](#) demonstrate how the implications of underlying productivity heterogeneity can be summarized by “reduced” import demand systems and that those demand systems can be identified from a cross-section of bilateral trade observations as long as the underlying system is assumed to be identical across countries.³ Since one microfoundation for this case occurs when all countries draw productivities independently from the same marginal distribution, the bounds we compute

²In this regard, our analysis also builds on the vast literature on gravity models of trade synthesized by [Costinot and Rodríguez-Clare \(2014\)](#) and [Head and Mayer \(2013\)](#).

³An advantage of the reduced demand system approach in [Adao et al. \(2017\)](#) is that it summarizes the joint role played by technology, relative factor endowments, and tastes for generating trade. Our analysis focuses on a single factor setting and holds preferences constant across all models considered so it only relates to the technology side of their analysis.

in the special case where we impose this i.i.d. restriction explores what is possible along the direction proposed by [Adao et al. \(2017\)](#).

Turning to monopolistic competition, a wide range of studies have introduced models where countries' productivities continue to be drawn independently, as in the gravity model, but from marginal distributions that depart from the case of common Pareto. For example, [Helpman et al. \(2004\)](#) examine country-specific Pareto distributions; [Helpman et al. \(2008\)](#), [Melitz and Redding \(2015\)](#), and [Feenstra \(2018\)](#) consider the truncated Pareto version; [Head et al. \(2014\)](#), [Bas et al. \(2017\)](#), and [Fernandes et al. \(2023\)](#) analyze log-normal distributions; [Nigai \(2017\)](#) proposes a mixture of log-normal and Pareto; and [Erhardt et al. \(2021\)](#) and [Adao et al. \(2026\)](#) allow for nonparametric distributions estimated separately by country.⁴ The primary contribution of our analysis is to generalize this exploration to encompass all possible distributions. We also highlight the (unavoidable) extrapolation that occurs when mapping from firm-level micro-data—an option in the monopolistic competition model case, since each product corresponds to a unique firm—to the welfare cost of autarky, since that cost depends only on the shape of a country's productivity distribution over a range of support for which no firm is active in the trading equilibrium. Our analysis therefore bounds all possible versions of this extrapolation.

2 Quantitative trade model environments

We begin by describing our core economic environment: a multi-country, single-factor trade model with many goods and arbitrary distributions of productivities (across goods and countries) around the world. We then describe how one can conduct counterfactual exercises (such as computing the gains from trade for any country) in such a model. Finally, we describe the version of this model that obtains when the global productivity distribution takes a particular form, as in workhorse versions of the model. Our presentation of these elements begins with a competitive version of this model (i.e. a Ricardian model of international trade) but then turns to the monopolistically competitive version (i.e. a [Melitz \(2003\)](#) model of international trade).

2.1 A competitive model of trade

Setup. There are D countries, potentially trading any of the goods along a continuum indexed by $\omega \in [0, 1]$. Each country d produces goods using a factor of production that is

⁴Several of these studies also follow [Eaton et al. \(2011\)](#) in allowing for additional forms of production-side heterogeneity beyond that in [Melitz \(2003\)](#), such as heterogeneity in fixed costs of exporting in [Fernandes et al. \(2023\)](#) and [Adao et al. \(2026\)](#). Our analysis sticks to the [Melitz \(2003\)](#) case.

owned by a representative household there and is immobile across countries.

Demand. The representative household in each country d has CES preferences across the continuum of goods:

$$U_d = \left(\int_0^1 q_d(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}},$$

with σ capturing the elasticity of substitution and $q_d(\omega)$ denoting the quantity of good ω consumed in country d . As is standard, we work with the case in which goods are substitutes (i.e. $\sigma > 1$).⁵

We let $p_{od}(\omega)$ denote the price of good ω in country d , if it were to be shipped from country o to country d . The determination of these prices involves the supply side of the economy, introduced below. But regardless of such determination, utility-maximization implies that the expenditure in d on good ω from country o will be

$$X_{od}(\omega) = E_d \gamma_d^{-1} p_{od}(\omega)^{1-\sigma}, \quad (1)$$

where E_d is total expenditure in d and where

$$\gamma_d \equiv \sum_o \int_0^1 p_{od}(\omega)^{1-\sigma} d\omega, \quad (2)$$

is a transformation of the standard CES price index. In particular, the real income of the representative household in d is given by $W_d = E_d \gamma_d^{\frac{1}{\sigma-1}}$.

Supply. The core of the Ricardian model lies on the supply side. We assume that producers in country o are endowed with a technology with productivity $z_o(\omega)$ for producing any good ω . The marginal cost of a producer of ω in o when selling in country d is given by

$$MC_{od}(\omega) \equiv \frac{w_o(1+t_{od})}{A_{od}z_o(\omega)}, \quad (3)$$

where w_o is the cost of a unit of the factor of production in o , t_{od} denotes the ad-valorem tariff charged in d on imports from o , and A_{od} is a bilateral productivity shifter.⁶ The productivity level $z_o(\omega)$ augments production by shifting downwards the marginal cost

⁵This restriction is needed to ensure that the gains from trade, discussed below, always remain finite as we consider alternative technologies in which d may be unable to produce some goods.

⁶The shifter A_{od} incorporates (inversely) an “iceberg” transport cost that acts like a tariff but generates no tax revenue.

of producing ω in o .

Because the good ω is homogeneous, perfect competition implies that prices in location d are determined by the marginal cost of the lowest-cost supplier to that location. In particular, the price of the version of good ω that hails from o and is sold in d must satisfy

$$p_{od}(\omega) = \begin{cases} MC_{od}(\omega) & \text{if } o = \arg \min_{o' \in D} MC_{o'd}(\omega), \\ \infty & \text{otherwise.} \end{cases} \quad (4)$$

Naturally, consumers in country d only buy from the minimum-price supplier of any good; that is, $p_{od}(\omega) = \infty$ implies $X_{od}(\omega) = 0$ in equation (1).

Equilibrium. We let L_o denote the (exogenous) supply of the factor of production in o , meaning that total factor income in o is given by $w_o L_o$. We assume that tariff revenue is spent on projects that do not directly contribute to utility.⁷ To account for trade imbalances, we define country d 's trade deficit as the difference between expenditure and income $B_d \equiv E_d - w_d L_d$. A competitive global equilibrium in this model is then one in which the factor market clears,

$$w_o L_o = \sum_d \int_0^1 \frac{1}{1 + t_{od}} X_{od}(\omega) d\omega \quad (5)$$

in every country o .

2.2 Stochastic representation

The key ingredient of this competitive model is the joint distribution of countries' productivities, $z_o(\omega)$ for all $o \in D$. For example, if productivities were to be distributed in such a way that they are perfectly correlated across countries then no country would have a comparative advantage in any good and there would be no trade (or gains from trade). On the other hand, if countries' distributions were relatively uncorrelated, or even negatively correlated, then a rationale for international specialization and hence trade emerges in the competitive equilibrium.

To emphasize the role of such distributions we follow the approach in [Eaton and Kortum \(2002\)](#) and treat the productivity $z_o(\omega)$ as a random variable that is drawn from a multi-country stochastic process. However, in a departure from such work we allow for

⁷This assumption simplifies the formulas presented below, which makes our core focus—the role of productivity heterogeneity—clearer. In practice, the share of tariff revenue in expenditure is negligible for the countries in our application.

an *arbitrary* joint cumulative distribution function F . That is, F denotes a function $F(\{z\})$ that equals the probability that the D -dimensional vector of random variables $\{z_o(\omega)\}$ lies below the D -dimensional vector $\{z\}$. Because F is the joint distribution across countries this notation allows for two important possibilities: that countries have different marginal distributions (e.g. that country o tends to get more dispersed draws than country o' does) and that countries' draws display forms of dependence (e.g. that the draws of country o are negatively correlated with those of country o').

Using this stochastic representation we let the random variable $p_{od}(\omega)$ be defined as a function of $z_o(\omega)$ using equations (3) and (4). Similarly, the random variable $X_{od}(\omega)$ is defined, via equation (1), as a function of $p_{od}(\omega)$. Finally, we follow the law-of-large-numbers convention of using the expectation operator $\mathbb{E}_F[\cdot]$ over the random variable $\{z_o(\omega)\}$ instead of integrals across the continuum of goods. Doing so, we can re-write the endogenous variable γ_d from equation (2) as

$$\gamma_d = \sum_o \mathbb{E}_F \left[p_{od}(\omega)^{1-\sigma} \right], \quad (6)$$

and the market clearing condition (5) as

$$w_o L_o = \sum_d \frac{1}{1 + t_{od}} \mathbb{E}_F [X_{od}(\omega)]. \quad (7)$$

For future reference, we denote the total value of bilateral trade as $X_{od} \equiv \mathbb{E}_F [X_{od}(\omega)]$ and the total bilateral import share in expenditure as $\lambda_{od} \equiv X_{od}/E_d$.

2.3 Counterfactuals

Our goal is to study how economic conditions in any given country d respond to counterfactual changes in the exogenous primitives of this model. In particular, we consider a change from the initial set of tariffs $\{t\}$ to the counterfactual set $\{t'\}$, and do so while holding constant all other exogenous model features—endowments $\{L\}$, bilateral productivity shifters $\{A\}$, preferences σ , the deficit to GDP ratio $\{B/Y\}$, and the multi-country productivity distribution F .

The counterfactual equilibrium is one in which all endogenous variables have changed correspondingly. This means that each producer's counterfactual marginal cost is $MC'_{od}(\omega) = w'_o(1 + t'_{od})A_{od}^{-1}[z_o(\omega)]^{-1}$ and the counterfactual price $p'_{od}(\omega)$ follows from (4) but eval-

uated at $MC'_{od}(\omega)$. Similarly, counterfactual expenditures are

$$X'_{od}(\omega) = (w'_d L_d + B'_d)(\gamma'_d)^{-1} p'_{od}(\omega)^{1-\sigma}, \quad (8)$$

$$\text{with } \gamma'_d = \sum_o \mathbb{E}_F [p'_{od}(\omega)^{1-\sigma}], \quad (9)$$

and equilibrium factor prices clear markets in the counterfactual equilibrium

$$w'_o L_o = \sum_d \frac{1}{1 + t'_{od}} \mathbb{E}_F [X'_{od}(\omega)]. \quad (10)$$

Gains from trade. A key counterfactual exercise of interest concerns the gains from trade for country d . This is calculated by changing tariffs from their factual values of $\{t\}$ to the counterfactual tariffs $\{t'\}$ that would place d in autarky (i.e. $t_{od} = \infty$ for all $o \neq d$) and then solving for the proportional change in real income. Following literature conventions, we define country d 's gains from trade GT_d as the change in its real income (factual minus autarky counterfactual) divided by its factual value, or

$$GT_d \equiv 1 - \left(\frac{w'_d L_d + B'_d}{w_d L_d + B_d} \right) \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}}. \quad (11)$$

This value is (minus) the percentage change in country d 's real income that would occur if its tariffs were to change from their factual values of $\{t\}$ to autarky levels $\{t'\}$.

Given the numeraire choice of $w_d = w'_d = 1$, and using the convention that d 's deficit is fixed as a share of its nominal GDP, the expression in (11) further simplifies to

$$GT_d = 1 - \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}}. \quad (12)$$

Though our focus in this paper will be on gains from trade, additional sets of counterfactual calculations are also feasible.⁸

⁸In particular, the methods regarding finite-neighborhood bounds described below can be applied to solve for sharp upper and lower bounds on any counterfactual calculation that takes the form $\kappa = \mathbb{E}_F[k(\{z(\omega)\}, \{w\}, \{w'\}, \{\gamma\}, \{\gamma'\}, \{t\}, \{t'\}, \{L\}, \{L'\}, \{A\}, \sigma)]$ for any known function $k(\cdot)$. This includes the effects of expanded sets of shocks (such as trade liberalization, technological change, and endowment growth) on expanded sets outcomes (such as trade flows, wages, terms of trade, etc.).

2.4 The Eaton and Kortum (2002) competitive model

Our presentation of the competitive Ricardian model so far has held for any given distribution of global productivities F . However, one particularly tractable version of F , proposed in the seminal model of Eaton and Kortum (2002), is that where countries' productivity levels are drawn independently from a Frechet distribution with scale parameter equal to one and shape parameter equal to $\theta^* > 0$, i.e. $z_o(\omega) \sim \text{Frechet}(1, \theta^*)$ for all o .⁹ We refer to this distribution as F^* in what follows (for the subset of our analysis referring to the competitive model) and treat it as a key reference point with which to compare the competitive model based on other distributions F .

We note that F^* imposes three separate types of restrictions relative to the unrestricted F : (i) countries' distributions are independent from one another, so the joint global distribution is the product of country-specific marginal distributions; (ii) those country-specific marginal distributions are all common; and (iii) the common marginal distribution takes the particular form of $\text{Frechet}(1, \theta^*)$.

2.5 Monopolistic competition and the Melitz (2003) model

Having described the general Ricardian model of trade under perfect competition, along with its Eaton and Kortum (2002) special case, we turn now to the case of monopolistic competition a la Melitz (2003). This is similar in many respects to the Ricardian model, so our discussion is brief.

Model setup. This model continues to feature D countries in which the representative household has CES preferences and hence demand follows equation (1). Production continues to be supplied by producers with marginal costs $MC_{od}(\omega)$ given by equation (3), on the basis of (variable) productivity draws $z_o(\omega)$ from an *arbitrary* global distribution that we continue to denote by F .

However, there are two differences relative to the Ricardian model. First, market conduct is now monopolistically (rather than perfectly) competitive, and there are fixed costs of producing. Monopolistically competitive conduct implies (in this CES demand case) that if a producer from o sells good ω in d then the price $p_{od}(\omega)$ will be marked up over the producer's marginal cost by the amount $\sigma/(\sigma - 1)$. In addition, a bilateral fixed cost

⁹That is, $F^*(\{z\}) = \prod_o F_o^*(z)$, where $F_o^*(z) = \Pr(z_o(\omega) \leq z) = e^{-z^{-\theta^*}}$. We note that some authors, including Eaton and Kortum (2002), define $F_o^*(z)$ to include a scale parameter $T_o > 0$ that is o -specific, with $F_o^*(z) = e^{-T_o z^{-\theta^*}}$, rather than $T_o = 1$ as in our convention. Because we include the free parameters A_{od} in (3), and these parameters enter the model symmetrically to $z_o(\omega)$, our choice of $T_o = 1$ is without loss.

$w_o f_{od}$ is incurred whenever a producer from o sells in d , so production will only take place where variable profits exceed such fixed costs.¹⁰ Collecting these observations we have

$$p_{od}(\omega) = \begin{cases} \frac{\sigma}{\sigma-1} MC_{od}(\omega) & \text{if } X_{od}(\omega) > \sigma f_{od} w_o, \\ \infty & \text{otherwise.} \end{cases} \quad (13)$$

The inequality $X_{od}(\omega) > \sigma f_{od} w_o$ in (13) is equivalent to a condition that productivity exceeds an endogenous pair-specific cutoff, i.e., $z_o(\omega) > \hat{z}_{od}^*$. We impose two standard restrictions on the parameters that give rise to the baseline equilibrium. First, we assume that parameters are such that $\hat{z}_{od}^* \geq \hat{z}_{oo}^*$ for all $d \neq o$, which ensures that selection into exporting is always active in the baseline. Second, we assume that all baseline cutoffs are above the lower bound of the support of F .

The other main difference brought about by monopolistic competition is that the equilibrium mass N_o of potential producers in o who choose to take a productivity draw $z_o(\omega)$ from F is endogenous. We assume that taking such a draw costs $w_o f_{Eo}$ and that a free entry condition mandates that the global profits a producer from o will obtain, in expectation, after drawing its productivity will be equal to the cost $w_o f_{Eo}$. This implies:

$$\sum_d \mathbb{E}_F \left[\max \left\{ 0, \frac{1}{\sigma} X_{od}(\omega) - w_o f_{od} \right\} \right] = w_o f_{Eo}. \quad (14)$$

Since aggregate profits are zero, household income in o continues to be given by $Y_o = w_o L_o$.

The market clearing and price index expressions of (6) and (7) are similar to the competitive model, though now augmented to include N_o ,

$$\gamma_d = \sum_o N_o \mathbb{E}_F \left[p_{od}(\omega)^{1-\sigma} \right], \quad (15)$$

$$w_o L_o = \sum_d \left(\frac{\sigma-1}{\sigma} \frac{1}{1+t_{od}} + \frac{1}{\sigma} \right) N_o \mathbb{E}_F [X_{od}(\omega)]. \quad (16)$$

Finally, we note that, in this model, total bilateral trade is given by $X_{od} = N_o \mathbb{E}_F [X_{od}(\omega)]$.

Summarizing, equilibrium in the monopolistically competitive model corresponds to a vector of global wages $\{w\}$, price indices $\{\gamma\}$, and masses of potential producers $\{N\}$ such that equations (1), (3) and (13)-(16) hold for all countries.

¹⁰As is standard, we assume that, in equilibrium, firms produce differentiated products. This means that there is only one producer per good ω in any destination market d .

Gains from trade counterfactuals. Simulations of scenarios involving counterfactual tariffs $\{t'\}$, such as those that implement autarky for a given country of interest d , can be computed using a similar procedure to that discussed for the Ricardian model above. That is, we hold constant all exogenous primitives—endowments $\{L\}$, bilateral productivity shifters $\{A\}$, bilateral fixed costs of production $\{f\}$, entry costs $\{f_E\}$, preferences σ , and the global distribution of productivities F —and solve for the new equilibrium at counterfactual $\{t'\}$ in which all endogenous variables—wages $\{w\}$, price indices $\{\gamma\}$ and masses of potential firms $\{N\}$ —adjust to the new equilibrium. We then compute the gains from trade for any country d following again the formula in equation (12), where $\{t'\}$ implements autarky for country d .

Reference distribution from Chaney (2008). As with the Ricardian model, our presentation of the monopolistic competition model à la Melitz (2003) has allowed for the global distribution of productivity levels F to take any form. However, since the pioneering work of Chaney (2008), it has become common to deploy one particularly tractable distribution: that in which each country o draws its productivity $z_o(\omega)$ independently from a Pareto distribution with scale parameter equal to one and a shape parameter equal to $\theta^* > 0$.¹¹ We therefore take this to be the reference distribution, denoted F^* , when studying the monopolistic competition model. As with the Ricardian model, relative to the unrestricted distribution F , the reference distribution F^* imposes three separate types of restrictions: full independence, common marginals, and particular (i.e. $\text{Pareto}(1, \theta^*)$) marginals.

3 Estimating the gains from trade

Calculating the gains from trade GT_d for any country d requires estimates of all of the model's exogenous features. These features include the observed inputs (tariffs $\{t\}$ and endowments $\{L\}$). They also include the model's unobserved parameters, such as the demand elasticity σ and bilateral technology shifters $\{A\}$ that apply to both competitive and monopolistically competitive models, plus the additional parameters for bilateral fixed costs $\{f\}$ and entry costs $\{f_E\}$ that apply in the case of the monopolistic competition model. Finally, and perhaps most fundamentally of all, the model's set of exogenous

¹¹That is, $F^*(\{z\}) = \prod_o F_o^*(z)$, where $F_o^*(z) = \Pr(z_o(\omega) \leq z) = 1 - z^{-\theta^*}$. As with the competitive model explained in footnote 9, there is no loss of generality in setting the shape parameter (and hence lower support) of this distribution to one.

features includes the global distribution of productivities F that governs comparative advantage, trade possibilities, and hence the gains from trade.

3.1 Gravity models

Both of the special cases described in Section 2—the [Eaton and Kortum \(2002\)](#) version of the competitive model and the [Chaney \(2008\)](#) version of the monopolistic competition model—are distinctive because they work with a particular version F^* of the global productivity distribution F such that bilateral trade flows between countries take the so-called gravity form. Following [Arkolakis et al. \(2012\)](#) we therefore refer to these as gravity models.

In the case of gravity models, estimation of GT_d follows three well-known steps. Indeed, as shown by [Arkolakis et al. \(2012\)](#), when these three steps are executed on the same dataset the resulting value of GT_d will be identical for the [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) models. We therefore discuss the general approach that applies to both of these models simultaneously here.

Step 1: Estimate θ^* from an orthogonality assumption. The first step is to estimate the single unknown parameter inside F^* , referred to in both models as θ^* . This begins with the assumption that the model predictions for each of the D^2 bilateral trade flows X_{od} correspond to those in the data. Denoting values in the data by $\tilde{X}_{od}$, this amounts to imposing the data-matching constraints

$$E_{F^*}[X_{od}(\omega)] = \tilde{X}_{od} \quad \text{for all } o \text{ and } d. \quad (17)$$

Then, estimation of θ^* relies on the fact that gravity models predict that aggregate bilateral trade flows X_{od} will follow the gravity equation:

$$\ln X_{od} = \alpha_o + \alpha_d - \theta^* \ln(1 + t_{od}) + \varepsilon_{od}. \quad (18)$$

Here, α_o and α_d collect model terms—including endogenous ones such as the wage of the origin and the income of the destination—that are common to a given origin and destination, respectively; and ε_{od} collects all exogenous model terms that are bilateral, other than the tariffs t_{od} . For example, in the [Eaton and Kortum \(2002\)](#) case, we have $\alpha_o = -\theta^* \ln w_o$, $\alpha_d = \ln E_d - \ln \gamma_d$, and $\varepsilon_{od} = \theta^* \ln(A_{od})$. The [Chaney \(2008\)](#) case is similar but with added considerations for the endogenous number of varieties and bilateral fixed costs: $\alpha_o = \ln N_o + \left(1 - \frac{\sigma}{\sigma-1}\theta^*\right) \ln w_o$ and $\varepsilon_{od} = \theta^* \ln A_{od} + \left(1 - \frac{\theta^*}{\sigma-1}\right) \ln f_{od}$.

A standard approach to estimating the parameter θ^* therefore involves using data on observed bilateral trade flows $\tilde{X}_{od}$ and ad-valorem bilateral tariffs t_{od} in order to estimate equation (18) by OLS. One way to partial-out the endogenous terms α_o and α_d is to double-difference equation (18) with respect to sample means. Letting, for example, $\Delta_o y_{od} \equiv y_{od} - \frac{1}{D} \sum_d y_{od}$ for any variable y_{od} , this procedure amounts to an OLS regression of $\Delta_o \Delta_d \ln \tilde{X}_{od}$ on $\Delta_o \Delta_d \ln(1 + t_{od})$.¹² The coefficient from this regression will provide a consistent estimate of $-\theta^*$ when the following orthogonality condition holds:

$$\sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \varepsilon_{od} = 0. \quad (19)$$

In the case of the [Eaton and Kortum \(2002\)](#) model the orthogonality condition (19) requires the assumption that tariffs are orthogonal to bilateral productivity shifters in the following sense

$$\sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \ln A_{od} = 0. \quad (20)$$

Similarly, consistent estimation for the [Chaney \(2008\)](#) case follows under the assumption that bilateral fixed costs are also orthogonal to tariffs:

$$\sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \ln f_{od} = 0. \quad (21)$$

Step 2: Estimate other model parameters by matching all trade flows. Armed with an estimate of θ^* from step 1, the second step uses this to obtain estimates of the remaining model parameters of interest. Given the data-matching constraint in (17), each of the D^2 bilateral residuals in equation (18)—i.e. $\ln \tilde{X}_{od} + \theta^* \ln(1 + t_{od})$ —provides an unbiased estimate of each of the D^2 terms ε_{od} , up to two irrelevant o -specific and d -specific scale factors. These estimated values of ε_{od} then provide (along with θ^*) unbiased estimates of the remaining model parameters that are necessary for counterfactual simulations (such as computing GT_d). For example, in the [Eaton and Kortum \(2002\)](#) model, since $\varepsilon_{od} = \theta^* \ln(A_{od})$, the translation from each bilateral residual to estimates of each bilateral productivity parameter A_{od} is immediate. Similarly, standard arguments in the [Chaney \(2008\)](#) case imply that the same combination of A_{od} and f_{od} that is identified from ε_{od} is sufficient for all equilibrium calculations. Finally, a remaining unknown model parameter in both cases is σ but in both models this turns out to only affect the scale of consumer welfare, so it cancels out of GT_d .

¹²In cases with no missing observations this is algebraically identical to an OLS regression of $\ln X_{od}$ on $\ln(1 + t_{od})$ and origin and destination fixed effects.

Step 3: Solve for counterfactual changes in the estimated model. The final step is to solve for GT_d , as defined in equation (12), according to the estimated model.¹³ In the case of the [Eaton and Kortum \(2002\)](#) version of the gravity model, this amounts to solving the system of equations (6) and (7) for the factual equilibrium and (9) and (10) for the counterfactual one, each time with the expectations $\mathbb{E}_F[\cdot]$ evaluated at $F = F^*$ (and where F^* corresponds to independent draws from a *Frechet*(1, θ^*) distribution at the estimated value of θ^* obtained from step 1). Similarly, in the [Chaney \(2008\)](#) case, step 3 requires solving equations (14)-(16) for the factual equilibrium, and their counterfactual equilibrium analogs, while setting the distribution F^* to be independent draws from *Pareto*(1, θ^*) at the estimated value of θ^* . Either way, as discussed previously, the [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) models are empirically isomorphic—so whenever steps 1 and 2 have been executed on the same trade and tariff data, these two models return the same value of GT_d . Indeed, as [Arkolakis et al. \(2012\)](#) have shown, under F^* the result of step 3 for gravity models is given simply by the formula

$$GT_d^{\text{ACR}} \equiv 1 - \lambda_{dd}^{\frac{1}{\theta^*}},$$

where $\lambda_{dd} \equiv X_{dd}/w_d L_d$ is the share of d 's expenditure at baseline that it sources from itself.

3.2 Beyond gravity models: distribution-agnostic bounds

We have seen how estimation of GT_d works in the case of the [Eaton and Kortum \(2002\)](#) version of the Ricardian model and the [Chaney \(2008\)](#) version of the monopolistic competition model. That is, in either case, we have seen how to estimate GT_d for one particular version, denoted F^* , of the global productivity distribution F . This raises two natural questions: How could the three-step estimation procedure of Section 3.1 be repeated for alternative distributions $F \neq F^*$? And how would GT_d change as these alternative distributions are considered?

One direction would be to repeat the analogs of steps 1-3 for particular alternative distributions. For example, one could imagine working with the Ricardian case and positing that the global distribution F corresponds to independent draws from a log-normal marginal distribution, with location parameter zero and scale parameter s , that is common to all countries. This would no longer be a gravity model. But the assumption that log tariffs $\ln(1 + t_{od})$ are mean independent of the log bilateral productivity shifter $\ln A_{od}$

¹³The second and third steps of this procedure are often combined into an algebraically equivalent single step via the “exact-hat” algebra procedure used by [Dekle et al. \(2009\)](#).

could nevertheless be incorporated, via a simulated method of moments procedure, to estimate s and all of the values of A_{od} (up to scale) as residuals. Having done that, the model could be solved—evaluating the expectations $\mathbb{E}_F[\cdot]$ by simulation, since they are not available in closed form for this choice of F —and GT_d could be calculated for this particular version of the global productivity distribution.

A concern with this sort of exercise is that it (unavoidably) starts with a particular choice of F , and that choice may be difficult to motivate. In this section we therefore characterize the bounds on gains from trade that could obtain across any choice of F , and refer to these as distribution-agnostic bounds. We begin with the Ricardian case.

Bounds on GT_d in the competitive model. We seek to estimate sharp bounds on the size of country d 's gains from trade GT_d that are consistent with the same three-step estimation method used for gravity models in Section 3.1, but now done for all productivity distributions F rather than for the particular F^* assumed by Eaton and Kortum (2002).

To do so, we proceed as follows. Consider the upper bound on GT_d denoted by $\overline{GT}_d$. This is given by the solution to the following program, where the counterfactual tariffs $\{t'\}$ are those that would place d in autarky (i.e. $t_{od} = \infty$ for all $o \neq d$):

$$\begin{aligned} \overline{GT}_d \equiv & \max_{F, \{A\}, \{w\}, \{w'\}, \{\gamma\}, \{\gamma'\}} 1 - \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}} \\ \text{subject to: } & \begin{cases} \text{equilibrium } \{w\} \text{ and } \{\gamma\} \text{ hold at } \{t\}, \\ \text{equilibrium } \{w'\} \text{ and } \{\gamma'\} \text{ hold at } \{t'\}, \\ \mathbb{E}_F[X_{od}(\omega)] = \tilde{X}_{od} \text{ for all } o \text{ and } d, \\ \text{and } \sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \ln A_{od} = 0. \end{cases} \end{aligned} \quad (22)$$

That is, estimating $\overline{GT}_d$ amounts to finding the version of the Ricardian model whose productivity distribution F is chosen to maximize the drop in welfare for country d when it moves from the initial equilibrium to autarky as per equation (12), while ensuring that: (i) factual wages $\{w\}$ and price indices $\{\gamma\}$ satisfy the factual equilibrium equations (1), (3), (4), (6) and (7); (ii) counterfactual wages $\{w'\}$ and price indices $\{\gamma'\}$ satisfy the counterfactual equilibrium equations (8)-(10); (iii) each model-implied bilateral trade flow in the initial equilibrium matches that in the data $\tilde{X}_{od}$; and (iv) model-consistent bilateral productivity shifters $\{A\}$ are orthogonal to tariffs in the sense of equation (20). The analogous lower-bound, which we denote by $\underline{GT}_d$, involves an identical program to that in (22) but with the min operator instead. The following proposition describes these upper

and lower bounds.

Proposition 1(a) (Bounds for the competitive model). *Suppose that the baseline data on trade shares satisfy, for country d of interest, the condition: $\lambda_{dd}/\lambda_{dn} \geq \lambda_{od}/\lambda_{on}$ for all $o, n \neq d$. Consider all possible productivity distributions F . Then the smallest and largest possible values for d 's gains from trade in the competitive model, as defined in program (22), are given by*

$$\underline{GT}_d = 0 \quad \text{and} \quad \overline{GT}_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (23)$$

The proof of this and all other propositions is in Appendix B. In all cases our proofs are constructive so they provide both a given GT_d bound (such as either of those in 23) and an example of a distribution that achieves it. In this sense the agnostic bounds we provide are always sharp.

Before turning to the intuition behind Proposition 1(a), we comment on the data condition in its premise. This condition is easily verified in any given dataset and is satisfied for all countries in our application. This is the case because the diagonals of the trade share matrix (i.e. λ_{dd} , which describe countries' purchases from themselves) tend to be much larger than other elements in the matrix because most countries are not very open to trade.

Returning to the intuition, a first step is to note that one can combine (i) the definition of λ_{dd} , (ii) the fact that autarky requires $\gamma'_d = \mathbb{E}_F[MC_{dd}(\omega)^{1-\sigma}]$, and (iii) the choice of numeraire ($w_d = w'_d = 1$), to write equation (12) as

$$GT_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}} \left(1 + \frac{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbb{1}\{d \neq \arg \min_o MC_{od}(\omega)\}]}{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbb{1}\{d = \arg \min_o MC_{od}(\omega)\}]} \right)^{\frac{1}{\sigma-1}}. \quad (24)$$

Consider the ratio of expectations in (24). This is the ratio of country d 's productivity for goods that it imports to the country's productivity on goods that it sells domestically. Since productivity $z_d(\omega)$ is weakly positive, the smallest value this ratio can take is zero, in which case we obtain the upper bound of $1 - \lambda_{dd}^{\frac{1}{\sigma-1}}$ in Proposition 1(a). This highlights how a necessary feature of any distribution F that can achieve the upper bound is that it puts a mass point with zero productivity for the goods that d imports at baseline.

A remaining challenge is to find such a distribution that is capable of matching the constraints in program (22)—generating an equilibrium that matches all baseline trade flow values and the orthogonality condition (20). The core content of Proposition 1(a) is to show formally that the set of distributions F is flexible enough—through both its marginals and its dependence structure—to achieve this. Our specific construction starts

with the distribution F^* which, as discussed in Section 3.1, is able to match all required constraints. We then remove from this distribution all the mass from the goods that d imports to zero productivity and then rescale it in a way that continues to satisfy all constraints. As we show, this is only possible when the data condition in the premise of Proposition 1(a) is satisfied. We also show—for this and all the bounds discussed in this section—that an F can be constructed for which all countries, not just d , see their gains from trade simultaneously at the level of the upper bound in (23). That is, country d 's gain can be maximized without precluding the possibility that other countries also achieve their own maximal levels.

Turning to the lower bound in Proposition 1(a), an equivalent way of writing (24) is

$$GT_d = 1 - \left(\frac{\mathbb{E}_F[MC_{dd}(\omega)^{1-\sigma}]}{\mathbb{E}_F[(\min_o MC_{od}(\omega))^{1-\sigma}]} \right)^{\frac{1}{\sigma-1}}. \quad (25)$$

Seen this way, GT_d would achieve the minimal value of zero if the distribution of productivities F could be such that the expected marginal cost from domestic production is the same, on average, as that from the cheapest global seller to d . This occurs in the opposite type of scenario from the upper bound case: it requires that country d can produce all imported goods at exactly the same tariff-inclusive prices as occur at baseline. In such a scenario, baseline imports merely reflect indifference, so shutting down trade need not raise the price index at all and gains from trade can be zero. Again, we construct an example that achieves this by starting from F^* , adding mass to the segments necessary for such indifference to hold, and then rescaling subject to the constraints in (22). As we show, this is again only possible when the data condition in the premise of Proposition 1(a) is satisfied.

Bounds on GT_d in the monopolistic competition model. Having characterized the competitive model case, the problem of obtaining distribution-agnostic bounds for the monopolistic competition model proceeds similarly. That is, we seek to bound GT_d for the Melitz (2003) model across all productivity distributions F rather than the particular version F^* that was used in the benchmark gravity model case of Chaney (2008). Yet we do so while replicating the standard three-step estimation procedure, which ensures that all bilateral trade flow data points are matched by the model at any of the distributions F under consideration, as well as that bilateral productivity shifters and fixed costs are orthogonal to tariffs in the sense of equations (19) and (21).

For the sake of interpretability we impose two restrictions on the set of Melitz (2003) models that we evaluate here. First, we require that the counterfactual of interest does

not change the mass of entrants, i.e. $N_o = N'_o$ for all o . This choice is conservative from the perspective of maximizing differences relative to the F^* Pareto case because N_o is always fixed in the F^* model.¹⁴ Second, we require that all firms in country d produce in autarky—i.e., that the endogenous cutoff in autarky, $(\hat{z}_{dd})'$, is equal to the lower bound of the support of F (which we normalize to one, without loss). This is also conservative because, in the absence of such a restriction, an optimizer could always change the bilateral fixed cost parameters $\{f\}$ so as to mimic a change in the support of F without actually changing F .

The program for maximizing GT_d in the monopolistic competition model is similar to that in (22), but now written as

$$\begin{aligned} \overline{GT}_d \equiv & \max_{F, \{A\}, \{f\}, \{f_E\}, \{w\}, \{w'\}, \{\gamma\}, \{\gamma'\}, \{N\}} 1 - \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}} \\ \text{subject to: } & \begin{cases} \text{equilibrium } \{w\}, \{\gamma\} \text{ and } \{N\} \text{ hold at } \{t\}, \\ \text{equilibrium } \{w'\}, \{\gamma'\} \text{ and } \{N\} \text{ hold at } \{t'\}, \\ N_o \mathbb{E}_F[X_{od}(\omega)] = \tilde{X}_{od} \text{ for all } o \text{ and } d, \\ \sum_o \sum_d \Delta_o \Delta_d \ln A_{od} \times \Delta_o \Delta_d \ln(1 + t_{od}) = 0, \\ \sum_o \sum_d \Delta_o \Delta_d \ln f_{od} \times \Delta_o \Delta_d \ln(1 + t_{od}) = 0, \\ \text{and } (\hat{z}_{dd})' = 1. \end{cases} \end{aligned} \quad (26)$$

Relative to the Ricardian model, the new considerations of the monopolistically competitive model naturally require an augmented set of model equations, as laid out in Section 2.5. It also requires a larger set of model parameters, due to the addition of bilateral fixed costs of production $\{f\}$ and entry costs $\{f_E\}$. In addition, the equilibrium number of potentially producing firms $\{N\}$ must be solved for at every candidate distribution F under consideration alongside the factual and counterfactual market-clearing wages $\{w\}$ and $\{w'\}$ and price indices $\{\gamma\}$ and $\{\gamma'\}$. Finally, we append the orthogonality condition for bilateral fixed costs (21) and the condition that all firms are active in autarky (i.e. $(\hat{z}_{dd})' = 1$).

The following proposition describes how the bounds $\underline{GT}_d$ and $\overline{GT}_d$ in the monopolistic competition model turn out to be identical (though under a different data condition) to those in the competitive model.

Proposition 1(b) (Bounds for the monopolistic competition model). *Suppose that the baseline tariff data satisfy the condition that there exists at least one pair of countries, $k_1 \neq d$ and k_2 ,*

¹⁴That is, under F^* we have $N_o = \frac{\sigma-1}{\sigma\theta} \frac{L_o}{f_{E_o}}$, which is a constant.

with $k_1 \neq k_2$, such that $\Delta_o \Delta_d \ln(1 + t_{k_1 k_2}) > 0$. Consider all possible productivity distributions F . Then the smallest and largest possible values for d 's gains from trade in the monopolistic competition model, as defined in program (26), are given by

$$\underline{GT}_d = 0 \quad \text{and} \quad \overline{GT}_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (27)$$

Just as with the Ricardian model above, the constructions we offer for achieving these bounds require that a data condition—stated in the premise of Proposition 1(b)—is satisfied. In this case the condition involves only tariff data and is extremely mild. Indeed, it holds generically whenever $D \geq 3$, as in our application.

The intuition for the monopolistically competitive case is similar to that seen above and, as before, constructive arguments are contained in Appendix B. The key distinction relative to the competitive model is that each good is now a distinct variety globally. Country d 's gains from trade can therefore achieve the bounds in Proposition 1(b) by adjusting the distribution of productivities among the potential producers in d that will begin production in autarky so that they are either highly productive (which minimizes GT_d) or unproductive (which maximizes it). This construction would be trivial, were it not for the need to ensure that such a distribution can satisfy the constraints in program (26). Our contribution is to establish that this is possible, provided that the stated tariff data condition holds.

3.3 Beyond gravity models: distributions “close” to gravity

Our analysis has so far developed bounds on the gains from trade that are possible if one allows for any productivity distribution F . As we discuss below, these bounds are many times wider than those obtained under the distribution F^* assumed by gravity models. To explore this gulf further we now describe a procedure for estimating sharp on GT_d that hold for neighborhoods of distributions F that are “close” to the reference distribution F^* . Varying the size of the neighborhood will permit a systematic investigation of the possibilities as one considers larger and larger departures from the workhorse case at F^* .

Neighborhoods of the F^* distribution. To begin, we use a standard method for comparing two distributions from the statistics literature. In particular, the “ ϕ -divergence” (henceforth divergence) $D(F||F^*) \geq 0$ of F from F^* is defined as

$$D(F||F^*) \equiv \int_{\{z\} \in \mathbb{R}_D^+} \phi \left(\frac{dF(\{z\})}{dF^*(\{z\})} \right) dF^*(\{z\}), \quad (28)$$

for a suitable convex function $\phi(\cdot)$. The form in $D(F||F^*)$ applies the penalty function $\phi(\cdot)$ to deviations of F from F^* at each point $\{z\}$ in the support of F^* and then integrates over all such penalties, weighted according to the distribution F^* . Several conventional choices of $\phi(\cdot)$ are available, such as the Kullback-Leibler (KL) form of $\phi(x) \equiv x \ln x - x + 1$ and the χ^2 form of $\phi(x) \equiv (x - e)^2 / (2e) + (x - e) + 1$. The particular definition of divergence that we use below is the “hybrid” divergence, recommended by [Christensen and Connault \(2023\)](#) for cases like ours where F^* has fat tails; this uses KL divergence for all $x \leq e$ and χ^2 divergence for $x > e$.¹⁵ Based on this definition of divergence, we define the neighborhood $\mathcal{N}_\delta$ of distributions around F^* as the set of all distributions F for which $D(F||F^*) \leq \delta$.

There are three attractive features of divergence for this purpose. First, it does not depend on the choice of units for the random variable $z_o(\omega)$. Second, transformations of location and/or scale in F^* do not affect divergence. Finally, as $\delta \rightarrow \infty$, any distribution that is absolutely continuous with respect to F^* —that is, any distribution that can be obtained by reweighting the baseline distribution within its support—can be approximated arbitrarily well by distributions contained within $\mathcal{N}_\delta$. Since the [Eaton and Kortum \(2002\)](#) version of F^* has full support over positive values, and the [Chaney \(2008\)](#) version has full support above a value (of one) that is set without loss since it is the Pareto distribution’s scale parameter, this means that all continuous distributions are considered as $\delta \rightarrow \infty$.

Armed with the previous definition of a neighborhood $\mathcal{N}_\delta$ of distributions F around F^* we define the gains from trade bounds within such a neighborhood as

$$\overline{GT}_d(\delta) \equiv \max_F \overline{GT}_d \quad \text{s.t.} \quad F \in \mathcal{N}_\delta. \quad (29)$$

That is, the finite-neighborhood bound $\overline{GT}_d(\delta)$ is simply the same optimization problem as earlier—program (22) in the competitive case and (26) in the monopolistically competitive case—but subject to the added constraint that the only admissible productivity distributions F must lie are within divergence δ of F^* . This means that the analytical bounds in Section 3.2 are approached in the $\delta \rightarrow \infty$ limit: $\overline{GT}_d = \overline{GT}_d(\delta = \infty)$ and $\underline{GT}_d = \underline{GT}_d(\delta = \infty)$.

While the program in (29) is straightforward to state, obtaining its solution is far from trivial. For any $\delta > 0$ the set of distributions in $\mathcal{N}_\delta$ to be considered is infinite, so conducting a search in the space of distributions F is clearly infeasible. And it is unclear how one

¹⁵The dual formulation in [Christensen and Connault \(2023\)](#) is valid under χ^2 divergence if, for any constraint $\mathbb{E}[g(z(\omega))]$ in (22) and (26), the moment function $g(z(\omega))$ satisfies $\mathbb{E}_{F^*}[g(z(\omega))^2] < \infty$. We show in Appendix A that, in both the Ricardian and monopolistically competitive models, this condition is equivalent to $\theta^* > 2(\sigma - 1)$, which is easily satisfied in our data.

could obtain analytical derivations like those of Section 3.2 for this finite-neighborhood case. Fortunately, however, Christensen and Connault (2023) have recently developed tools for making computational progress on problems of this sort, showing that a particular sub-problem of (29) has a dual that is feasible. In particular, their result refers to cases that involve maximizing, over all distributions F in $\mathcal{N}_\delta$, an objective function that can be written as (a known function of) an expectation over F subject to any number of constraints that can also be written as (known functions of) expectations over F . In such a case, Christensen and Connault (2023) establish that the dual of this infinite-dimensional program is concave and finite-dimensional, with dimensionality given by the number of constraints, and where all expectations in this dual program are expectations over F^* .

The program in (29) does not immediately fit this desired form where the objective function and all constraints involve only expectations over F . For example, the choice set in $\overline{GT}_d$, as defined in equation (22) for the competitive model case, directly involves model parameters such as the productivity shifters, $\{A\}$. Similarly, the factual and counterfactual equilibrium objects $(\{w\}, \{\gamma\})$ and $(\{w'\}, \{\gamma'\})$ enter the objective function (12) and constraints in nonlinear ways that disrupt the pure expectations form. To circumvent these challenges, Christensen and Connault (2023) propose that one treat all such variables—not just primitives such as $\{A\}$, but also endogenous variables such as $\{w\}$, $\{w'\}$, $\{\gamma\}$ and $\{\gamma'\}$ in our context—as an expanded set of “parameters”, which we denote ψ . After doing so, the program in (22) can be split into two stages. First, for any given value of the parameter vector ψ , solve the maximization program over $F \in \mathcal{N}_\delta$ by converting it from an infeasible infinite-dimensional program into a feasible finite-dimensional program. Second, optimize over ψ in an outer loop.

We execute this nested two-step procedure below. But the large dimensionality of our problem—with $D^2 + 4D$ parameters in the outer loop of (22) and $2D^2 + 6D$ in that of (26)—motivated our exploitation of several dimensionality-reduction steps that apply to our problem that we discuss next.

3.4 Implementation Details

We implement this two-step optimization program using Knitro, where an inner loop solves the search over distributions for a given vector of parameters, and an outer loop searches over the vector of parameters. We compute expectations using Monte Carlo integration with 800,000 draws.

In both versions of the model, there are a large number of parameters in the outer loop (in the order of at least D^2), which would be very challenging for any solver for all but a

trivial number of countries D . However, we can substantially reduce this dimensionality by exploiting the fact that many of these parameters are irrelevant for computing gains from trade for d . For the Ricardian case, the parameters $A_{od'}$ for $d' \neq d$ do not appear in any equations relevant for determining GT_d , which means they can be omitted.¹⁶ The only exception is that they do appear in the moment restriction corresponding to the gravity equation, discussed in Section 3.3, so we instead implement this moment using the subset of observations where $d' = d$.¹⁷ The same logic allows us to omit $\{\gamma_o, \gamma'_o, w'_o\}_{o \neq d}$. Since baseline trade flows imply a unique (to scale) vector of wages, $\{w_o\}$, we can solve for these directly. We can also normalize $w_d = w'_d = 1$ as well as one element of $\{A_{od}\}_o$. Finally, we can omit equation (6) as it is directly implied by the full set of trade flow moments. This means we are left with a parameter vector that consists of $\{\gamma_d, \gamma'_d, \{A_{od}\}_{o \neq 1}\}$, which is of dimension $D + 1$.

The Melitz (2003) case is similar, and actually benefits from similar dimensionality-reduction even though in its original form it is higher-dimensional than the Ricardian model. In addition to omitting $A_{od'}$ for $d' \neq d$ and $o \neq d$, we can also omit $\{A_{od}\}_{o \neq d}$. This is feasible because the constraint $N_d \mathbb{E}_F[X_{dd}(\omega)] = \tilde{X}_{dd}$ —which can never be omitted as it is necessary to identify A_{dd} and thus γ'_d —has no direct dependence on other countries, except through d -level aggregate objects (such as γ_d and N_d) that are treated as parameters.¹⁸ Moreover, combining the zero-profit condition (14) and the market clearing condition (16) allows us to eliminate all $A_{dd'}$ for $d' \neq d$ as aggregate profits are zero and these variables are not directly needed to determine d 's consumption. These observations imply it is sufficient to aggregate all other countries $d' \neq d$ and reduce the overall set of parameters to $\{\gamma_d, \gamma'_d, A_{dd}, f_{dd}, f_{Ed}, A_{dx}, f_{dx}\}$, where A_{dx} and f_{dx} are auxiliary parameters that represent the technology shifter and the fixed costs in the aggregated export market with trade flow $X_{dx} \equiv w_d L_d - X_{dd}$. As a result, this case has only 7 parameters in the outer loop.

¹⁶We also then omit the corresponding trade share constraints, $\tilde{X}_{od} = \mathbb{E}_F[X_{od'}(\omega)]$ for $d' \neq d$. This does not affect the model's ability to replicate the baseline equilibrium, as for any combination of F and the other parameters, it is possible to find a vector $\{A_{od'}\}_{o, d' \neq d}$ that can rationalize the omitted trade shares. The dimensionality reduction comes from the observation that we do not need to know the values of these omitted A_{od} to compute our counterfactual of interest.

¹⁷As this removes the ability to include origin fixed effects, the moment is equivalent to an instrumental variables estimation of $\ln \tilde{X}_{od}$ on $\ln w_o(1 + t_{od})$, using $\ln(1 + t_{od})$ as an instrument.

¹⁸This is not possible in the Ricardian case because the constraint $\mathbb{E}_F[X_{dd}(\omega)] = \tilde{X}_{dd}$ contains all $A_{od'}$ for $d' = d$ in the min function of equation (4).

3.5 Data

Our analysis focuses on data from 2016. We obtain: aggregate bilateral trade data $\tilde{X}_{od}$ from the OECD ICIO database; tariff data t_{od} from TRAINS, constructing a measure of the average bilateral tariffs by weighting 6-digit HS-level tariffs by their corresponding flows from the CEPII BACI database; and total labor force data L_o from the World Bank's WDI database. In order to keep the dimensionality manageable, we work with a sample of countries that includes the 19 largest countries by total expenditure and aggregate all remaining countries into a Rest of World group. This results in $D = 20$.

4 Quantifying the sensitivity of quantitative trade models

We turn now to the following question: how sensitive are the conclusions that standard quantitative models make about the gains from trade to changes in the underlying distribution of productivities? For now, our focus lies on estimates of GT_d for the case where d is France, a country with import penetration that is representative among the countries in our sample.

4.1 Gains from trade in the gravity model

Workhorse quantitative trade models of the gravity form, such as [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#), predict how the welfare of a country d , such as France, would change if the country's tariffs were to rise to the point of autarky. These predictions about GT_d are made by employing the model estimation procedure detailed in Section 3.1 for a version of the model in which global production productivities F are drawn from the particular distribution F^* —the i.i.d. *Frechet* $(1, \theta^*)$ distribution in the case of [Eaton and Kortum \(2002\)](#) or the i.i.d. *Pareto* $(1, \theta^*)$ distribution in the case of [Chaney \(2008\)](#). Since the distribution F^* is, of course, a divergence $\delta = 0$ from itself, the [Arkolakis et al. \(2012\)](#) point estimate of GT_d obtained when assuming $F = F^*$ is the point at which our upper and lower bounds from Section 3.3 coincide, yielding

$$\overline{GT}_d(\delta = 0) = \underline{GT}_d(\delta = 0) = GT_d^{\text{ACR}} = 1 - \lambda_{dd}^{\frac{1}{\theta^*}}. \quad (30)$$

We replicate standard estimation procedures in order to obtain an estimate of GT_d^{ACR} for France in 2016. First, the OLS estimate of θ^* in equation (18) is 6.8. Second, in the 2016 data of our application, France has a value of $\lambda_{dd} = 0.835$ (i.e. the country imports 16.5% of its expenditure). Putting these two numbers together, our estimate of France's

gains from trade, according to the gravity model of either [Eaton and Kortum \(2002\)](#) or [Chaney \(2008\)](#), is $1 - (0.835)^{\frac{1}{6.8}} = 0.026$. In other words, according to the workhorse gravity model, aggregate welfare in France would drop by 2.6% if the country's trade policy were to retreat into autarky.

The remainder of our analysis is focused on exploring the sensitivity of this estimate to alternative distributions of global productivities.

4.2 Distribution-agnostic bounds on the gains from trade

We turn now to calculating the range of values for GT_d that are possible if one is completely agnostic about the distribution of productivities in either the competitive or monopolistically competitive models. Propositions [1\(a\)](#) and [1\(b\)](#) have established that, in both cases, and under data conditions that hold for all countries in our application, the lower bound is $\underline{GT}_d = 0$ and the upper bound is $\overline{GT}_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}$.

Like the [Arkolakis et al. \(2012\)](#) formula for $\overline{GT}_d(\delta = 0) = \underline{GT}_d(\delta = 0)$ in the gravity model case, the expression for the maximum possible value $\overline{GT}_d$ depends on the baseline value of λ_{dd} , discussed in Section [4.1](#) above. But rather than depending on the F^* shape parameter θ^* , the distribution-agnostic version $\overline{GT}_d$ depends on σ , the elasticity of substitution across products. Perhaps surprisingly, the value of σ does not enter the [Arkolakis et al. \(2012\)](#) formula at all, so the standard estimation template for GT_d in gravity models does not deliver (or require) an estimate of σ . But σ does matter for counterfactuals under essentially all other possible distributions $F \neq F^*$. We therefore work with the value of $\sigma = 2.5$ and explore robustness to this value in ongoing work.

Using this value of σ in the formula $\overline{GT}_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}$ implies that the maximal value that France's gains from trade can take, across all possible production heterogeneity distributions F , is 11.3% (i.e. $1 - (0.835)^{\frac{1}{2.5-1}}$). This value is 4.4 times larger than that of 2.6% predicted by the workhorse model under F^* . Combined with the fact that the distribution-agnostic lower bound is $\underline{GT}_d = 0$, we conclude that a very wide range of predictions about GT_d are possible if one were completely agnostic about the underlying production heterogeneity in either the competitive or monopolistically competitive model.

4.3 Interpreting measures of divergence δ

One reaction to the wide bounds $\underline{GT}_d$ and $\overline{GT}_d$ reported above is that the stance of complete agnosticism about productivity distributions is too nihilistic. We therefore re-

evaluate these bounds for a smaller set of distributions by narrowing the neighborhood size $\mathcal{N}_\delta$ such that δ is finite (but still greater than zero so that we consider departures from F^*). Before reporting on this exercise in Section 4.4 below we first pause to provide guidance on how a divergence of any given value, such as δ , can be interpreted.

To provide one benchmark, consider the sensitivity analysis conducted for the case of gravity models when the value of the shape parameter θ^* is adjusted, as is common practice among users of quantitative trade models. Begin with the [Eaton and Kortum \(2002\)](#) case, where the global productivity distribution F^* implies that $z_o(\omega)$ is distributed, across o and ω , such that each country o draws its own $z_o(\omega)$ independently from the distribution $\text{Frechet}(1, \theta^*)$. Now consider changing this distribution to an alternative, denoted $F^{\text{alt.}}(\theta)$, that remains an [Eaton and Kortum \(2002\)](#) gravity model but with a different shape parameter, θ . That is, $F^{\text{alt.}}(\theta)$ is analogous to F^* but based on $\text{Frechet}(1, \theta)$ at some $\theta \neq \theta^*$. Because this distribution $F^{\text{alt.}}(\theta)$ differs from F^* we can calculate the resulting divergence, $D(F^{\text{alt.}}(\theta) || F^*)$.

The red line in Figure 1 plots the value of $D(F^{\text{alt.}}(\theta) || F^*)$ against θ for this Frechet case estimated in Section 4.1.¹⁹ That is, F^* corresponds to $D = 20$ countries taking independent draws from $\text{Frechet}(1, \theta^* = 6.8)$. Similarly, the blue line reports an analogous exercise for the case of [Chaney \(2008\)](#), where F^* is based on a $\text{Pareto}(1, \theta^* = 6.8)$ distribution and $F^{\text{alt.}}(\theta)$ is based on $\text{Pareto}(1, \theta)$. Naturally, in both cases the divergence is zero at $\theta = \theta^* = 6.8$, and the divergence is strictly greater than zero for all other values of $\theta \neq \theta^*$.

Our interest lies in the extent to which the divergence grows as θ differs increasingly from θ^* . For example, we see that divergence $D(F^{\text{alt.}}(\theta) || F^*)$ below the value of 2 corresponds to values of θ that span the range $\theta = [5.2, 9.8]$ in the Frechet case and $\theta = [4.5, 11]$ in the Pareto case. These windows seem well within the range of conventional sensitivity analysis that is already a key part of quantitative trade modeling.²⁰ We therefore view a neighborhood such as $\mathcal{N}_\delta$ at $\delta = 2$ —that is, the set of all distributions F that lie within $D(F || F^*) \leq 2$ of F^* —to be of substantial interest when conducting sensitivity analysis.

¹⁹The calculations in Figure 1 use the KL divergence rather than the Hybrid divergence used in all further analysis below. This is conservative (because KL divergence is always weakly smaller than Hybrid divergence) and avoids computation of D -dimensional numerical integrals at each value of θ (since KL divergence is available in closed form for the two cases calculated here).

²⁰For example, [Eaton and Kortum \(2002\)](#) deploy three procedures for estimating θ^* and arrive at values ranging from 3.60 to 12.86.

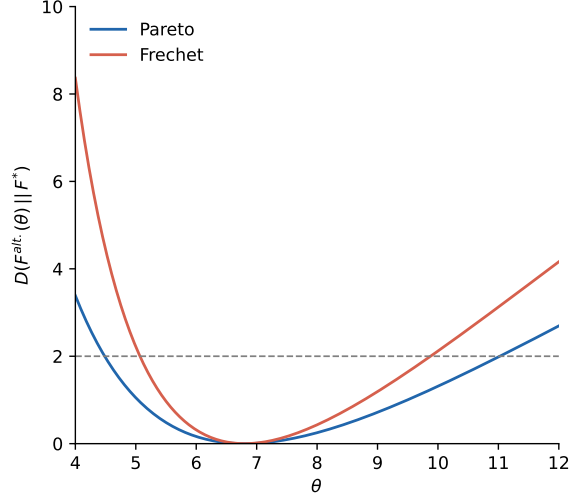


Figure 1: Examples of divergence between distributions

4.4 Gains from trade for distributions “close” to the gravity model

We now estimate bounds on counterfactual predictions about the gains from trade that hold for a varying set of neighborhoods $\mathcal{N}_\delta$ that are defined by their productivity distribution F lying within a divergence of δ from the gravity model distribution F^* . We confine attention to $\delta \leq 2$ because, as discussed in Section 4.3, this amount of divergence appears to contain distributions that are sufficiently “close” to the gravity model to be of interest.

As described in Section 3.3, for any given value of δ , we calculate the upper and lower bounds on country d ’s gains from trade, $\overline{GT}_d(\delta)$ and $\underline{GT}_d(\delta)$ respectively, that would be possible across all productivity distributions with divergence less than δ from the gravity model distribution F^* . And we do so while always ensuring that, given any distribution under consideration, it is fed into an otherwise standard equilibrium model that exactly matches all trade flow data points at the factual equilibrium as well as the standard orthogonality conditions of gravity model estimation. That is, we solve program (29) at the chosen value of δ , with $\overline{GT}_d$ drawing on program (22) in the competitive case and program (26) in the monopolistically competitive case; analogously, we obtain the lower bounds $\underline{GT}_d(\delta)$ by using a version of these same programs that minimizes GT_d rather than maximizing it. Throughout, we focus on the case where $d = \text{France}$.

Figure 2 plots our baseline results. For purposes of comparison, the dashed orange line displays the value of GT_d^{ACR} , the extent of gains from trade in the workhorse model (i.e. with $\delta = 0$); as discussed in Section 4.1, we estimate this to be 0.026. In addition, the dashed black lines report the distribution-agnostic upper and lower bounds, $\overline{GT}_d$ and $\underline{GT}_d$ respectively, that are possible across all distributions; as discussed in Section 4.2, we

estimate these bounds to be $\overline{GT}_d = 0.113$ and $\underline{GT}_d = 0$. Finally, we add to Figure 2 the values of $\overline{GT}_d(\delta)$ and $\underline{GT}_d(\delta)$ as a function of δ for the case of the Ricardian model (in solid blue) and the Melitz (2003) model (in solid red), each computed for five values of δ at each bound. As expected, the finite- δ bounds begin at the standard gravity model estimate of $GT_d = 0.026$ when δ is small and then approach the distribution-agnostic bounds $\overline{GT}_d$ and $\underline{GT}_d$ as δ grows larger.

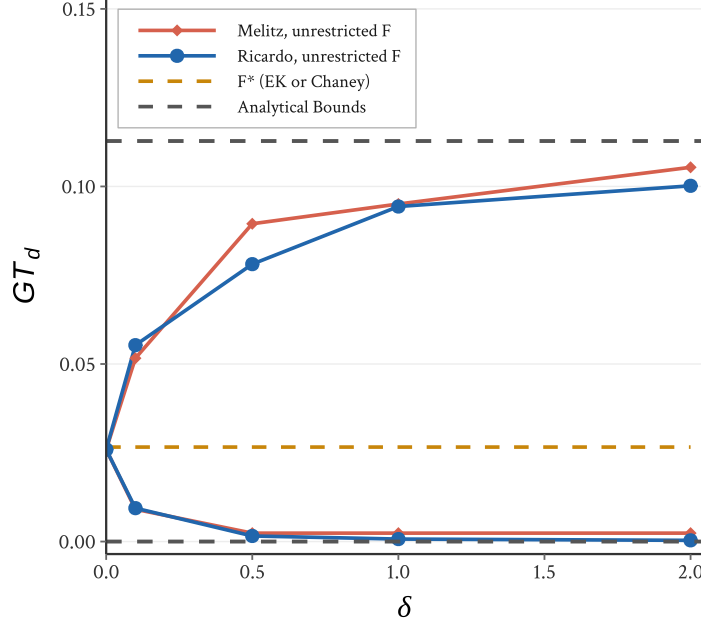


Figure 2: Bounds on gains from trade (for France) across alternative productivity distributions

An object of substantial interest is the *rate* at which the bounds diverge as δ is increased, starting at zero. This encapsulates a notion of sensitivity to alternative productivity distributions, and Figure 2 reveals a striking amount of sensitivity. For example, even at $\delta = 1$ the trade data support Ricardian models with gains from trade that are four times larger than in the gravity model, as well as Melitz (2003) models that are also four times larger. And at $\delta = 2$ —a value of divergence that, as discussed in Section 4.3, we view as corresponding to a conventional exploration of robustness in gravity models—the upper bounds in Figure 2 are very close (88% of the way for Ricardo and 92% for Melitz (2003)) to the distribution-agnostic maximal upper bound that is itself about 4.4 times larger than the gravity model gains.

4.5 Taking stock

Put together, the findings of this section can be summarized as follows. First, as a benchmark, the gravity model, with its productivity distribution F set to the workhorse version of F^* , and when estimated in the standard manner, predicts that French living standards would fall by 2.6% if that country were to cease trading internationally. Second, our new analytical bound formula implies that if one were agnostic about the productivity distribution F to use but followed exactly the same estimation steps as are applied to the gravity model then the cost to France of moving to autarky would lie between 0% and 11.3% of baseline real income. Finally, we find a similar range of values when we restrict the set of possible distributions F to only those that lie “close” to F^* (for example, via the restriction that the divergence of F from F^* must be no more than two). All of these conclusions apply to both the competitive version of the gravity model and the monopolistically competitive version.

One implication of these findings is that even the plain-vanilla trade model we study—one without features such as multiple sectors or input-output linkages that tend to increase gains from trade—is entirely consistent with the possibility that gains from trade are many times larger than they are when the gravity assumption of F^* distributions is imposed. But versions of the model with no gains from trade are possible too.

A second implication of our findings is that quantitative trade models appear sensitive to relatively small changes in their underlying productivity distributions. Starting from the workhorse distributional assumption of F^* we have seen how there exist distributions F that are “close” to F^* and yet deliver gains from trade that are much larger or smaller than is possible in the workhorse version. Indeed, almost all of what is possible in general, with complete agnosticism about distributions, is possible even for the set of distributions F that are “close” to F^* . One might conjecture that matching trade data and standard orthogonality conditions would ensure robustness of quantitative trade models to their underlying assumptions about productivity distributions. According to our analysis this is not the case.

5 High-level Restrictions on Productivity Distributions

Our main finding so far is that gains from trade can be many times larger or smaller than typically found in gravity models if one is agnostic about the model’s underlying productivity distribution F , and this remains true even when the set of admissible distributions is restricted to lie within a small-divergence neighborhood of the distribution used in the

gravity model F^* . In this section we evaluate how much our gains from trade bounds would shrink when the set of distributions under consideration is restricted—both from the starting point of agnosticism and from a given neighborhood around F^* —by requiring that distributions satisfy a range of high-level properties.

In particular, the high-level restrictions that we impose are motivated by three core features of the global productivity distributions F^* used by [Eaton and Kortum \(2002\)](#) in the competitive model and [Chaney \(2008\)](#) in the monopolistically competitive model. The first such feature is that each country takes independent draws—that is, there is no cross-country dependence or correlation across their draws. A second feature is that every country draws from the same marginal distribution. Finally, the particular marginal distribution that all countries draw from is $Frechet(1, \theta^*)$ in the [Eaton and Kortum \(2002\)](#) case and $Pareto(1, \theta^*)$ in the [Chaney \(2008\)](#) case. In this section we impose versions of these high-level characteristics on the distributions F that are admissible when we evaluate bounds on the gains from trade. We work through these restrictions one by one and, where possible, in combination.

5.1 Restrictions: perfect vs. monopolistic competition

One starting observation concerns a broad distinction between the competitive Ricardian and monopolistically competitive [Melitz \(2003\)](#) models. The competitive model involves head-to-head competition, in every destination market d , over the worldwide producers of each good ω who are based in each of the D origin countries. When country d moves from its factual trading equilibrium to the counterfactual state of autarky, the supplier of every imported good will shift from the foreign supplier (who had won this head-to-head competition in the factual equilibrium) to a domestic supplier. The gains from trade therefore depend on the size of the foreign supplier’s victory. The stochastic properties of this victory depend on country d ’s marginal productivity distribution and the correlation of d ’s draws with the best of the $D - 1$ foreign draws, a distribution that itself depends on those $D - 1$ foreign countries’ marginal distributions and their dependence. In short, the gains in the Ricardian model clearly depend on properties of domestic marginals, foreign marginals, and the dependence among them. We explore these components below.

On the other hand, consider the [Melitz \(2003\)](#) model. In this case, competition is not head-to-head for any good; instead, every firm pays an entry cost in order to be the global monopolist in its own product. When country d moves to autarky it will lose access to all foreign imports and seek to replace them with domestic alternatives. But the lack of head-to-head competition means that what matters for the gains is how well these

newly produced domestic varieties can replace the aggregate contribution of the now-unavailable foreign imports, which is purely a function of the domestic distribution. That is, conditional on knowing the marginal distribution of productivities in country d the additional properties of F —such as the $D - 1$ foreign countries' marginal distributions or the cross-country dependence of productivity draws—has no direct bearing on GT_d . Given these points, there is little to be learned by imposing high-level restrictions on the Melitz model. All results in the following subsections therefore focus on the Ricardian case alone.

5.2 Imposing independence and zero correlation

As discussed, the F^* used by the [Eaton and Kortum \(2002\)](#) version of the Ricardian model assumes that, for any good ω , countries o take their productivity draws $z_o(\omega)$ independently from one another. By contrast, the distributions F that we have considered so far have allowed for arbitrary forms of cross-country dependence. We now seek to narrow this gap between F^* and F by restricting the set of distributions F considered such that any admissible F must satisfy cross-country independence just like F^* does. In this case the only distinction between F^* and F that remains is that due to the particular marginal distributions that each country draws from.

Distribution-agnostic bounds. We begin by extending the distribution-agnostic upper-bound argument of Proposition [1\(a\)](#) to the case where distributions F must obey independence. That is, we characterize the bound $\overline{GT}_d^{\text{indep.}}$ that obtains if program [\(22\)](#) is solved subject to the constraint that the choice variable F (the multi-country stochastic distribution) must feature cross-country independence. Indeed, we show that, provided a stronger data condition than that in Proposition [1\(a\)](#) holds, the bounds with and without independence imposed are the same: $\overline{GT}_d^{\text{indep.}} = \overline{GT}_d$. This is perhaps surprising given that the construction used in the Proposition [1\(a\)](#) case exploited a strong form of dependence across countries.

Proposition 2 (Bounds for competitive model with independence). *Suppose that the baseline tariff and trade data satisfy the following condition: letting Ξ denote a square matrix composed of elements Ξ_{ij} that are zero apart from the d th row with elements given by $\frac{1}{1-\sigma} \ln \frac{\lambda_{dj}}{1-\lambda_{dj}}$, we have $\text{Var}(\Delta_i \Delta_j \ln(1 + t_{ij})) > \text{Cov}(\Delta_i \Delta_j \Xi_{ij}, \Delta_i \Delta_j \ln(1 + t_{ij}))$. Consider all possible productivity distributions F under which countries draw their productivities independently. Then the largest possible value for country d 's gains from trade in the competitive model, as defined in program*

(22), is given by

$$\overline{GT}_d^{\text{indep.}} = \overline{GT}_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (31)$$

Further, this value is approached arbitrarily closely by a sequence of distributions that have Frechet marginals.

Like those seen earlier, this proposition presumes a data condition that can be easily checked on any dataset and is verified for all countries in our application. The proof of Proposition 2 provides a construction of a particular F that involves countries drawing independently from a distribution with (non-identical) Frechet marginals. The marginal for country d has a shape parameter θ that approaches $\sigma - 1$ from above, in which case that country's production is dominated by a small mass of highly productive goods, such that in autarky the contribution of other goods is negligible and hence the denominator in (24) becomes arbitrarily large. The marginals for all countries other than d are all common and set to be Frechet with shape parameter increasing from θ^* until the data-matching and orthogonality constraints in program (22) are satisfied. As we show, this is always possible as long as the data condition in the premise of Proposition 2 holds.²¹

However, as this construction makes clear, without access to dependence, the data-matching and orthogonality constraints mean that the gains from trade for one country are being maximized in ways that will tend to reduce other countries' gains. This contrasts with the case of Propositions 1(a) and 1(b) where all countries could simultaneously achieve the bound $\overline{GT}_d$.

Bounds for F in a finite neighborhood of F^* . Proposition 2 suggests that when one considers all distributions it is essentially without loss to impose independence when studying country d 's gains from trade. However, as the construction provided suggests, this may be a limiting result that may not be very accurate in any neighborhood that is "close" to F^* . We therefore turn to a quantitative exploration of the case where F must both satisfy cross-country independence and lie within the neighborhood $\mathcal{N}_\delta$.

Recall that two jointly distributed random variables are independent if and only if it is the case that their joint CDF is equal to the product of their two marginal CDFs. We approximate this restriction about CDFs by using a finite number of expectations at L^I specific quantiles of the relevant CDFs. That is, we say that F satisfies approximate

²¹Intuitively, for given trade data, the construction needs to balance the resulting negative slope between model-implied bilateral productivities A_{od} and tariffs for country d with the resulting positive slope for other countries in order to maintain overall orthogonality. The data condition stated in Proposition 2 is violated if d 's exports are so correlated with tariffs that the values of A_{od} required to rationalize the data as θ approaches $\sigma - 1$ are simply too negatively correlated with tariffs for this to be possible.

independence when

$$\mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{z}_l\}\mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}] = \mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{z}_l\}]\mathbb{E}_F[\mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}], \quad (32)$$

holds for all pairs of countries $o \neq o'$ and all L^I quantiles l , and where $\bar{z}_l$ denotes the value of the l th quantile of a distribution of interest.²² Since this constraint imposes the necessary and sufficient condition for independence at a finite number of points in the distribution, by letting L^I grow we can approximate increasingly well the case of true independence.²³

We impose this form of independence on F by adding the $D(D-1)L^I = 1900$ constraints embodied in (32) to the program (29) when computing the upper bound $\overline{GT}_d(\delta)$ and the lower bound $\underline{GT}_d(\delta)$. This continues to search over all distributions $F \in \mathcal{N}_\delta$ that deliver Ricardian models that are consistent with trade data and the standard orthogonality condition, but with the space of such distributions now confined to those that feature (approximate) independence of productivity draws across countries.

Figure 3 displays the results from implementing this independence restriction: blue lines show the unrestricted bounds from Figure 2 and the dark green lines show those that impose independence. In the case of the lower bound $\underline{GT}_d(\delta)$, these lines are essentially coincident, implying that the bounds are unperturbed by the requirement that F features cross-country independence. On the other hand, the upper bound $\overline{GT}_d(\delta)$ with independence is somewhat lower than that facing no restrictions. For example, at the $\delta = 2$ level, $\overline{GT}_d(\delta)$ is 0.086 when F must exhibit cross-country independence and 0.1 when F is unrestricted. This implies that the maximum possible gains from trade in France would be about 18% (i.e. $1 - (0.086 - 0.026)/(0.1 - 0.026)$) less sensitive to the underlying productivity distribution F if it were believed that countries draw their technologies independently. As anticipated, even though Propositions 1(a) and 2 imply that imposing an independence restriction has no “bite” at $\delta \rightarrow \infty$, the bite is non-negligible at $\delta = 2$.

Stopping short of full independence, one may wish to impose weaker restrictions on the dependence of productivity draws across countries. For example, a condition of zero correlation can be obtained by imposing the far simpler version of (32) based on first

²²In practice, we use the common marginals of F^* , i.e. $\text{Frechet}(1, \theta^*)$, for this purpose. This has the advantage of using quantile values that are spaced out in the same proportion to their distribution under F^* .

²³In practice, we set $L^I = 5$ due to the large number of pairwise combinations in (32).

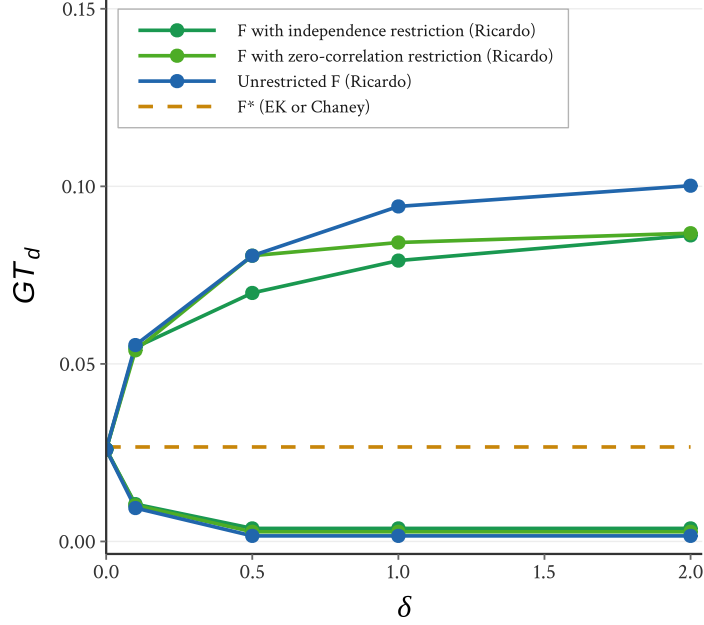


Figure 3: Ricardian gains from trade and cross-country dependence

moments alone:

$$\mathbb{E}_F[z_o(\omega)z_{o'}(\omega)] = \mathbb{E}_F[z_o(\omega)]\mathbb{E}_F[z_{o'}(\omega)] \quad \text{for all } o \neq o'. \quad (33)$$

We report on this zero-correlation case in the light green line shown in Figure 3. Naturally, since this restriction is weaker than that of full independence, the zero-correlation restriction bounds are weakly wider than the approximate independence restriction bounds (dark green lines). But the lines are close, especially for δ above 0.5. There is therefore not much of a distinction to be had from more complicated forms of dependence than linear correlation—a finding that is perhaps unsurprising, at least for high δ , given that even the strong restriction of full independence can achieve the maximal GT_d bound (Proposition 2).

These results are also informative for the prior literature that has relaxed the independence assumption in Eaton and Kortum (2002). For example, a special case of the analysis in Lind and Ramondo (2023) maintains the use of Frechet marginals but allows for non-zero amounts of correlation. Figure 3 illustrates an analogous idea but in the opposite direction: start with all distributions (blue lines) and then impose restrictions so as to keep only those with zero correlation (light green lines). We learn that this can reduce the maximal gains from trade, but not by much (only by 18% at $\delta = 2$). And it has essentially no effect on the minimal gains from trade.

5.3 Imposing common marginals

The second high-level restriction imposed in the [Eaton and Kortum \(2002\)](#) choice of F^* is that all countries o draw their productivities $z_o(\omega)$ from a common marginal distribution. Of course, the actual distribution F^* involves not only common marginals but specific marginals given by $\text{Frechet}(1, \theta^*)$. But the feature of commonality may derive from primitive assumptions on where technologies derive from, or simply from a desire for parsimony. So it is interesting to evaluate how the case of common (but flexible) marginals alone can shape gains from trade in data-consistent Ricardian models.

Distribution-agnostic bounds. We begin again with an extension of Proposition [1\(a\)](#) to the case where admissible distributions F must feature common marginals. In fact, the following proposition goes further and shows that, even with the additional restriction that countries' common marginals must all be $\text{Frechet}(1, \theta)$, the resulting upper bound on the gains from trade $\overline{GT}_d^{\text{CM}}$ will be the same as that of $\overline{GT}_d$ in Proposition [1\(a\)](#) as long as a stronger data condition holds. It is important to note, however, that the value of θ in this common Frechet marginal will differ, in general, from that of θ^* used in F^* .

Proposition 3 (Bounds for competitive model with common marginal distributions). *Suppose that the baseline tariff and trade data satisfy the data Condition [1](#) stated in Appendix [B.4](#). Consider all possible productivity distributions F under which countries draw their productivities from a common marginal distribution but with potential dependence across countries. Then the largest possible value for country d 's gains from trade in the competitive model, as defined in program [\(22\)](#), is given by*

$$\overline{GT}_d^{\text{CM}} = \overline{GT}_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (34)$$

Further, this bound is approached arbitrarily closely by a distribution with $\text{Frechet}(1, \theta)$ marginals.

The data condition behind this proposition is more involved than those seen earlier. This is unsurprising, given that stronger restrictions have been placed on the possible distributions that must satisfy all constraints despite achieving the same bound. While we relegate the exact condition to the Appendix, the rough idea is to partition all countries (other than d) into K groups in order to ensure that the sum across groups of certain within-group covariances (between tariffs and certain trade data summaries) is smaller than the overall variance of the tariff data. We find that this condition is satisfied for our application even when only $K = 2$ groups are considered.

The constructive proof of Proposition [3](#) uses a sequence of common Frechet distributions with shape parameter θ that approaches $\sigma - 1$ from above, but does so while using

the dependence structure to ensure that the orthogonality condition continues to hold. In the case of independent distributions (as in F^*) it is impossible for all countries to have a common Frechet marginal distribution with shape parameter different from θ^* , since this would violate the orthogonality condition (19). However, cross-country dependence creates more freedom. Our construction make draws correlated within blocks of countries, which (all else equal) behaves like a negative productivity shifter. The data condition ensures that one can construct blocks of countries with sufficiently high intra-block correlation between trade flows and tariffs that orthogonality can still hold overall despite the reduction in θ from θ^* .

Bounds in the neighborhood of F^* . Turning to the case of finite δ , like in the previous subsection we approximate a restriction about (marginal) CDFs by using a finite number of expectations at L^{CM} quantiles l . All countries' marginal distributions are approximately common if the following condition holds

$$\mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{z}_l\}] = \mathbb{E}_F[\mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}] \text{ for all } l \leq L^{CM} \text{ and } o' \neq o. \quad (35)$$

That is, common marginals across countries requires that the expected mass of the distribution below any given point $\bar{z}_l$ for one country o is the same as that for any other country o' .²⁴ The gains from trade bounds, such as $\overline{GT}_d(\delta)$, among all distributions F for which all countries share the same marginal distribution can then be approximated by adding the $(D - 1)L^{CM} = 950$ constraints embodied in (35) to the program (29).

While this method works arbitrarily well for large L^{CM} , one challenge in practice occurs in regards to the likelihood of the highest-productivity draws, which are particularly important for generating gains from trade. Since gains from trade depend on the price index, which is a transformation of the $(1 - \sigma)$ th moment of the price draws that consumers face, we mitigate the risks of inadequate upper-tail approximations by further imposing the additional $(D - 1)L^{CM}$ constraints

$$\mathbb{E}_F[z_o(\omega)^{(1-\sigma)} \mathbb{1}\{z_o(\omega) < \bar{z}_l\}] = \mathbb{E}_F[z_{o'}(\omega)^{(1-\sigma)} \mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}] \text{ for all } l \leq L^{CM} \text{ and } o' \neq o. \quad (36)$$

This ensures that the draws beneath each quantile have the same $(1 - \sigma)$ th moment.

The orange lines in Figure 4 report the results from this exercise. (Again, the blue

²⁴We again choose the points $\bar{z}_l$ as the L^{CM} quantiles of the $Frechet(1, \theta^*)$ distribution. For any given L^{CM} , the common marginals restriction in (35) requires far fewer restrictions than the pairwise restrictions required for independence in (32), so it is feasible to use a larger number of quantiles. We therefore set $L^{CM} = 50$ here.

lines display the unrestricted baseline case for comparison.) The message from this figure is that imposing the common marginals constraint does not do much to reduce the sensitivity of quantitative Ricardian models—not just at large δ as in Proposition 3, but even at small values of δ such as 1 or 2. In particular, the lower bound of zero is unaffected by this constraint and the upper bound is reduced, but only by about 11% (i.e. $1 - (0.092 - 0.026)/(0.1 - 0.026)$) at a value such as $\delta = 2$.

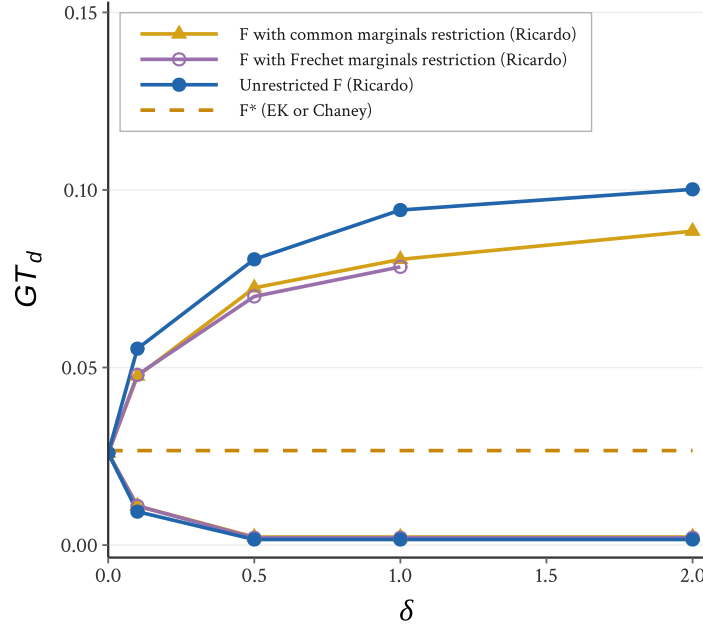


Figure 4: Ricardian gains from trade and countries' marginal distributions

5.4 Imposing common $Frechet(1, \theta^*)$ marginals

The previous restriction on F requires that all countries share the same marginal distribution, but it remains agnostic about what that marginal distribution is. A tighter restriction of interest is that the common marginal distribution takes a particular form, such as the $Frechet(1, \theta^*)$ distribution, as in the F^* used by Eaton and Kortum (2002).²⁵ We do this next. It is worth noting that in this case the only difference between F and F^* is that F remains agnostic about the cross-country dependence structure, whereas F^* assumes independence.

To see how this can be done, let $H(\cdot)$ denote the CDF of the inverse of the distribution $Frechet(1, \theta^*)$. Divide the unit-interval support of $H(\cdot)$ into $\bar{H}$ discrete and evenly-spaced

²⁵This is a distinct exercise from that in Proposition 2. Here we constrain the common marginal to be $Frechet(1, \theta^*)$, with θ^* set to the estimated value of 6.8; the construction in Proposition 3 instead uses common $Frechet(1, \theta)$ marginals with a flexible θ .

points indexed by h . Finally, let $\bar{H}_h$ denote the value of the h th quantile of $H(\cdot)$. Then we can ensure that the marginal of the distribution of productivity draws $z_o(\omega)$ in any country o has approximately the same marginal distribution as $\text{Frechet}(1, \theta^*)$, as desired, by matching each of these $\bar{H}$ quantile values, as long as $\bar{H}$ is large enough for this approximation to be accurate.²⁶ Formally, this amounts to imposing the constraint

$$\mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{H}_h\}] = H(\bar{H}_h) \quad \text{for all } o \text{ and all } h \leq \bar{H}. \quad (37)$$

In addition, for reasons discussed in Section 5.3, we ensure that the $(1 - \sigma)$ th moment of each quantile of every country's distribution is the same as that under $\text{Exp}(1)$:

$$\mathbb{E}_F[z_o(\omega)^{1-\sigma} \mathbb{1}\{z_o(\omega) < \bar{H}_h\}] = \mathbb{E}_{F^*}[z_o(\omega)^{1-\sigma} \mathbb{1}\{z_o(\omega) < \bar{H}_h\}] \quad \text{for all } o \text{ and all } h \leq \bar{H}. \quad (38)$$

Adding the $2D\bar{H} = 2000$ constraints in (37) and (38) to the program (29) ensures that bounds on GT_d at any δ only consider distributions F that deviate from F^* in terms of the cross-country dependence of their technology draws and not at all (at least approximately) in terms of the Frechet marginals that all countries share.

We report on this case in Figure 4. The purple lines impose the new restriction of common $\text{Frechet}(1, \theta^*)$ marginals. Again, counterfactual sensitivity remains substantial, despite requiring that all countries have the same marginal distributions as they do in Eaton and Kortum (2002). Perhaps surprisingly, the purple lines (where all marginals are the same and Frechet) are very close to the orange lines (where all marginals are the same but can be *any* marginal distribution). So, at least when dependence is left flexible, the sensitivity of the Ricardian model's predictions has little to do with the particular marginal distributions employed.

5.5 Imposing both independence and common marginals

The restrictions we have introduced so far have all been imposed one at a time, but it is of interest to also impose subsets of them simultaneously. One such combination is the case where F is *i.i.d.*—where countries draw productivities independently and from an identical marginal distribution. Indeed, when F is i.i.d, just as F^* is, the only remaining difference between F and F^* is that the marginals of F are flexible whereas those of F^* are $\text{Frechet}(1, \theta^*)$.

²⁶In practice we set $\bar{H} = 50$.

Distribution-agnostic bounds. The following proposition provides a sufficient condition under which the maximal gains from trade $\overline{GT}_d$ in Proposition 1(a) can be achieved even for the i.i.d. case.

Proposition 4 (Bounds for competitive model with i.i.d. distribution). *Suppose the base-line tariff and trade data satisfy Condition 2 in Appendix B.5. Consider all possible productivity distributions F under which countries draw their productivities independently from a common marginal distribution. Then the largest possible value for country d 's gains from trade in the competitive model, as defined in program (22), is given by*

$$\overline{GT}_d^{\text{IID}} = \overline{GT}_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (39)$$

The basic idea behind this proposition is to construct a distribution F whose identical marginals approach a distribution comprised of a mixture of two Dirac distributions at the points of zero and one, with an interior weight on each Dirac distribution that is chosen so as to match all countries' data points and satisfy the orthogonality condition. Data Condition 2 in Appendix B.5—which is verified for all countries in our sample other than Australia and Mexico—is a sufficient condition that ensures this is possible.

Bounds in the neighborhood of F^* . As before, even though Proposition 4 suggests that the imposition of the i.i.d. restriction will not shrink our estimated bounds across an infinitely large neighborhood of distributions, the construction offered in the proof that proposition suggests that the i.i.d. restriction may have substantial bite in finite neighborhoods of F^* . We therefore estimate the bounds for such a case in a manner analogous to that in Sections 5.2 and 5.3. That is, we add to the program (29) the stacked set of $D(D-1)L^I = 1900$ constraints in (32) that ensure approximate independence alongside the $2(D-1)L^{CM} = 1900$ constraints in (35) and (36) that ensure approximate common marginals.

The results of this exercise are shown in Figure 5. In particular, the light blue lines report on the case where all permitted distributions F satisfy both (approximate) independence and common marginals. For comparison, the dark blue lines again display the unrestricted case and the orange lines display that with common marginals (but not independence). Once again, the bounds conveyed by the light blue lines are still quite wide. Going from the unrestricted case to the doubly-restricted case of both independence and common marginals achieves, at $\delta = 2$ for example, a reduction in sensitivity of the upper bound by about 23.5% (i.e. $1 - (0.082 - 0.026) / (0.1 - 0.026)$). What is most surprising is that the combination of independence and common marginals (light blue in this figure)

achieves little more robustness in $\overline{GT}_d(\delta)$ than that of either independence (Figure 3) or common marginals (orange line in this figure) on their own.

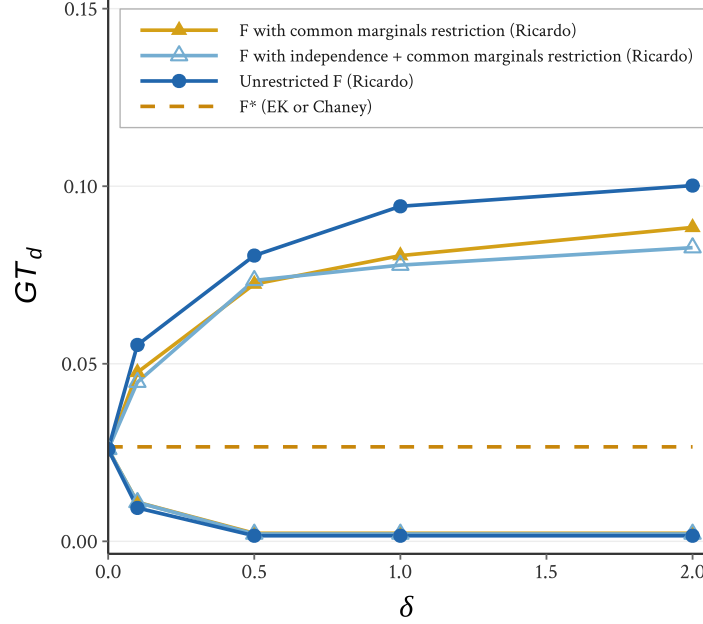


Figure 5: Ricardian gains from trade with independence and common marginals

This exercise is of particular interest for connecting with the empirical estimation of Ricardian models that are more flexible than the gravity version of [Eaton and Kortum \(2002\)](#). In particular, [Adao et al. \(2017\)](#) demonstrate that counterfactual predictions, such as those concerning the gains from trade, in nonparametric Ricardian models can be constructed from the knowledge of countries' import demand systems alone, and that such systems can be estimated using data on bilateral trade flows, tariffs, and exogeneity restrictions that generalize (19). With a single cross-section of data, as we have used here, import demand systems can only be estimated under the assumption that they are identical across countries. In the context of the competitive model in Section 2 this will obtain when F features exchangeability across countries, a condition that is implied by the stronger i.i.d. condition imposed here. That is, the similarity of the light and dark blue lines in Figure 5 gives a sense of how little is restricted when researchers assume that countries' import demand systems are identical.

5.6 Summary

Stepping back, the combined lesson from the restrictions imposed throughout this section is that gains from trade in the Ricardian model can be many times larger or smaller than

they are in the workhorse [Eaton and Kortum \(2002\)](#) version, even in the face of reasonable restrictions. Interestingly, our estimated bounds are wide when one departs from that model’s productivity distribution—namely, that countries’ productivities are drawn i.i.d. from $\text{Frechet}(1, \theta^*)$ marginals—in *either* of two high-level directions that are available. One such direction is to relax the independence assumption while maintaining that of common $\text{Frechet}(1, \theta^*)$ marginals, as we have explored in [Figure 4](#). The other direction is to relax the assumption of common $\text{Frechet}(1, \theta^*)$ marginals while maintaining the assumptions of independence and common marginals, as we have seen in [Figure 5](#). Either way, for the finite neighborhood within divergence $\delta = 2$ of the [Eaton and Kortum \(2002\)](#) distribution, we estimate bounds on gains from trade that are about 20% narrower than those for the completely unrestricted case but still contain Ricardian models with substantial (or tiny) gains from trade.

6 Concluding Remarks

Three modern revolutions have taken place in the study of international trade. A first is the desire to answer counterfactual questions using quantitative simulations. A second is the incorporation of micro-level production heterogeneity across goods and countries, either in Ricardian models a la [Eaton and Kortum \(2002\)](#) or monopolistically competitive models a la [Melitz \(2003\)](#). And a third is the desire to ensure that models exactly match all baseline trade flows, and their implied empirical patterns such as the fit of the gravity equation or the magnitude of correlation between tariffs and trade flows, before being used for counterfactual simulations away from the baseline scenario.

The goal of this paper has been to explore the robustness of this three-part revolution to variations in the assumed distribution of cross-country and cross-good heterogeneity in production technologies. Workhorse models such as [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) assume that this heterogeneity is drawn from particular distributions that have been chosen largely for their tractability and alignment with gravity model traditions. But given the impossibility of direct evidence about these distributions—since what matters for key counterfactual questions such as the gains from trade is the distribution of a country’s productivity among goods it does not produce—it seems important to probe the sensitivity of model predictions to variations in the underlying distribution of assumed production heterogeneity.

Our results convey a strong sense in which quantitative trade models are highly sensitive to the underlying distribution of production heterogeneity that is assumed. And this is true despite ensuring that every distribution of heterogeneity that we consider is

capable of generating, in full general equilibrium, the same values for all bilateral trade flows as occurs at the empirical baseline, as well as satisfying the same orthogonality conditions that are typically used for parameter estimation. Indeed, this finding continues to hold even when one maintains components of the workhorse distributions—e.g. that countries' draws are independent but with arbitrary country-specific marginals, or independent with arbitrary but common marginals, or with benchmark common marginals but arbitrary amounts of (positive or negative) dependence.

A natural direction for future work is to extend the methods in this paper to a wider set of model features. In principle, our model can be augmented to include elements such as domestic input-output linkages, international supply chains, multiple sectors, and multiple factors—all of which typically increase the gains from trade in quantitative models, implying that their omission here has been conservative. Likewise, phenomena such as factor mobility and consumer demand are typically modeled via distributional assumptions about the heterogeneity of households' preferences. The tools developed in this paper could be applied to such settings as well.

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A Derivations

A.1 Derivation of moment functions in monopolistically competitive model

Standard CES profit maximization implies

$$\begin{aligned} p_{od}(\omega) &= \frac{\sigma}{\sigma-1} MC_{od}(\omega) \\ X_{od}(\omega) &= \frac{E_d}{\gamma_d} \left(\frac{\sigma-1}{\sigma} \right)^{\sigma-1} MC_{od}(\omega)^{1-\sigma} \\ \pi_{od}(\omega) &= \frac{1}{\sigma} X_{od} \end{aligned}$$

A firm in o produces for market d if variable profits exceed the fixed of entering that markets, i.e., $X_{od}(\omega) > \sigma f_{od} w_o$. Some algebra implies that the cutoff is given by

$$z_{od}^* = \frac{\sigma}{\sigma-1} \frac{w_o(1+t_{od})}{A_{od}} \left(\frac{\gamma_d \sigma f_{od} w_o}{E_d} \right)^{\frac{1}{\sigma-1}}.$$

We require that parameters are such that this cutoff is in the interior of the support of F at baseline.

Aggregate post-entry profits are given by

$$\Pi_o = N_o \sum_d \mathbb{E}[(\pi_{od}(\omega) - f_{od} w_o) I_{od}(\omega)],$$

where $I_{od}(\omega) \equiv 1\{\pi_{od}(\omega) > f_{od} w_o\}$. We can rewrite this as

$$\Pi_o = N_o \sum_d \int_{z_{od}^*}^{\infty} \left(\frac{1}{\sigma} X_{od}(z) - w_o f_{od} \right) dF(z) = N_o \sum_d w_o f_{od} \int_{z_{od}^*}^{\infty} \left(\left(\frac{z}{z_{od}^*} \right)^{\sigma-1} - 1 \right) dF(z),$$

where the second equality uses the fact that cutoff z_{od}^* is defined as $X_{od}(z_{od}^*)/\sigma = w_o f_{od}$. Free entry implies that $\Pi_o = N_o f_{Eo} w_o$, which then gives the implicit definition of the cutoff from [Melitz and Redding \(2013\)](#):

$$f_{Eo} = \sum_d f_{od} \int_{z_{od}^*}^{\infty} \left(\left(\frac{z}{z_{od}^*} \right)^{\sigma-1} - 1 \right) dF(z). \quad (40)$$

A.1.1 Moments

The trade share moment matches $\lambda_d \equiv X_{od}/E_d$, where $X_{od} \equiv \mathbb{E}[X_{od}(\omega)1\{X_{od}(\omega) > \sigma w_o f_{od}\}]$. This is

$$\begin{aligned}\lambda_{od}(\omega) &= N_o \frac{p_{od}(\omega)^{1-\sigma}}{\gamma_d} 1\{p_{od}(\omega) < p_{od}^*\} \\ &= N_o \frac{1}{\gamma_d} \left(\frac{\sigma-1}{\sigma} \right)^{\sigma-1} \left(\frac{w_o(1+t_{od})}{A_{od}} \right)^{1-\sigma} z_o(\omega)^{\sigma-1} 1\{z_o(\omega) > z_{od}^*\}.\end{aligned}$$

The price index moment is

$$\begin{aligned}\gamma_d(\omega) &= \sum_o N_o p_{od}(\omega)^{1-\sigma} 1\{p_{od}(\omega) < p_{od}^*\} \\ &= \sum_o N_o \left(\frac{\sigma-1}{\sigma} \right)^{\sigma-1} \left(\frac{w_o(1+t_{od})}{A_{od}} \right)^{1-\sigma} z_o(\omega)^{\sigma-1} 1\{z_o(\omega) > z_{od}^*\} \\ &= \gamma_d \sum_o \lambda_{od}(\omega).\end{aligned}$$

The last line makes clear this this moment is redundant if we already match all of the trade shares.

For market clearing, given the formulation that tariffs represent real resources costs, total revenues must equal total factor income, which implies

$$w_o L_o = N_o \sum_d \mathbb{E}_F[X_{od}(\omega)1\{z_o(\omega) > z_{od}^*\}].$$

Since $\mathbb{E}_F[X_{od}(\omega)1\{z_o(\omega) > z_{od}^*\}] = \lambda_{od} E_d$, at baseline we can just solve directly for w_o and thus do not need to actually impose this moment.

A.1.2 Gains from Trade

We can show that $z_{dd}^* < z_{dd}^*$ as follows. Since Equation (40) must hold in both baseline and autarky, we have

$$f'_{dd}J((z_{dd}^*)') = f_{dd}J(z_{dd}^*) + \sum_{m \neq d} f_{dm}J(z_{dm}^*),$$

where $J(x) \equiv \int_x^\infty \left(\frac{z}{x}\right)^{\sigma-1} d(z)$. This implies $J((z_{od}^*)') > J(z_{od}^*)$, which in turn implies $(z_{od}^*)' < z_{od}^*$.

We can use the trade share moment for $o = d$ to obtain

$$\gamma_d = N_d \frac{1}{\lambda_{dd}} \left(\frac{\sigma - 1}{\sigma} \right)^{\sigma-1} \left(\frac{w_d(1 + t_{dd})}{A_{dd}} \right)^{1-\sigma} \mathbb{E}[z_d(\omega)^{\sigma-1} \mathbf{1}\{z_d(\omega) > z_{dd}^*\}].$$

Using the normalization that $w_d = w'_d = 1$ and $t_{dd} = 0$, we can then write gains from trade as

$$\begin{aligned} GT_d &= 1 - \frac{E'_d}{E_d} \left(\frac{\gamma'_d}{\gamma_d} \right)^{\frac{1}{\sigma-1}} \\ &= 1 - \lambda_{dd}^{\frac{1}{\sigma-1}} \left(\frac{E'_d}{E_d} \right) \left(\frac{N'_d}{N_d} \right)^{\frac{1}{\sigma-1}} \left(\frac{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbf{1}\{z_d(\omega) > (z_{dd}^*)'\}] }{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbf{1}\{z_d(\omega) > z_{dd}^*\}]} \right)^{\frac{1}{\sigma-1}} \\ &= 1 - \lambda_{dd}^{\frac{1}{\sigma-1}} \left(\frac{E'_d}{E_d} \right) \left(\frac{N'_d}{N_d} \right)^{\frac{1}{\sigma-1}} \left(1 + \frac{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbf{1}\{z_{dd}^* > z_d(\omega) > (z_{dd}^*)'\}]}{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbf{1}\{z_d(\omega) > z_{dd}^*\}]} \right)^{\frac{1}{\sigma-1}}. \end{aligned}$$

The first term captures changes in consumer income. The second term captures changes in entry. The third term captures changes in the average productivity of domestic goods.

A.1.3 Pareto

If $F(z) = 1 - z^{-\theta^*}$, then we can solve for z_{od}^* in closed-form and expenditure takes the usual gravity form,

$$X_{od} = N_o \int_{z_{od}^*}^{\infty} X_{od}(\omega) dF(\omega) = N_o \left(\frac{E_d}{\gamma_d} \right)^{\frac{\theta^*}{\sigma-1}} \left(\frac{\sigma - 1}{\sigma} \right)^{\theta^*} \left(\frac{A_{od}}{w_o(1 + t_{od})} \right)^{\theta^*} (\sigma f_{od} w_o)^{1 - \frac{\theta^*}{\sigma-1}} \frac{\theta^*}{\theta^* - (\sigma - 1)},$$

where the elasticity on $1 + t_{od}$ is $-\theta^*$.

A.2 Derivation of technical condition for convex duality

Proposition 2.1 in [Christensen and Connault \(2023\)](#) requires that any function $g(z(\omega))$ such that $\mathbb{E}_F[g(z(\omega))]$ appears in the programs (22) and (26), either as a constraint or in the objective, must satisfy a regularity condition that depends on the choice of divergence. For χ^2 divergence, the requirement is that $\mathbb{E}_{F^*}[g(z(\omega))^2] < \infty$. We derive the implications of this condition here under both the Ricardian and monopolistically competitive models. In both cases, it is sufficient to focus on the price index moment, as this is larger than all of the trade share moments.

A.2.1 Ricardian Model

The price index moment is $p_d(\omega)^{1-\sigma}$, where $p_d(\omega) \equiv \min_m p_{md}(\omega)$. Expressing this as a function of z :

$$\begin{aligned} p_d(\omega)^{1-\sigma} &= \left(\min_m p_{md}(\omega) \right)^{1-\sigma} \\ &= \left(\underbrace{\max_m \frac{A_{md}}{w_m(1+t_{md})} z_m(\omega)}_{\equiv z_d(\omega)} \right)^{\sigma-1}. \end{aligned}$$

When F^* is *Frechet*(1, θ^*), standard derivations imply that the distribution of this object is

$$\begin{aligned} \Pr(z_d(\omega) < z) &= \Pr(z_{md}(\omega) < z, \text{ for all } m) \\ &= \prod_m \exp \left(- \left(\frac{w_m(1+t_{md})}{A_{md}} z_m(\omega) \right)^{-\theta^*} \right) \\ &= e^{-\Phi_d z^{-\theta^*}}, \end{aligned}$$

where $\Phi_d \equiv \sum_m \left(\frac{w_m(1+t_{md})}{A_{md}} \right)^{-\theta^*}$.

We can therefore write the regularity condition as

$$\begin{aligned} \mathbb{E}_{F^*}[(p_d(\omega)^{1-\sigma})^2] &= \mathbb{E}_{F^*}[(z_d(\omega)^{\sigma-1})^2] \\ &= \int_0^\infty z^{2(\sigma-1)} \theta^* \Phi_d z^{-\theta^*-1} e^{-\Phi_d z^{-\theta^*}} dz \\ &= \Phi_d^{\frac{\theta^*}{2(\sigma-1)}} \int_0^\infty t^{\left(1 - \frac{2(\sigma-1)}{\theta^*}\right)-1} e^{-t} dt, \end{aligned}$$

where $t \equiv \Phi_d z^{-\theta^*}$. The integral converges (to $\Gamma(1 - \frac{2(\sigma-1)}{\theta^*})$) if $1 - \frac{2(\sigma-1)}{\theta^*} > 0$, or $\theta^* > 2(\sigma-1)$.

Monopolistic Competition

The regularity condition is

$$\mathbb{E}_{F^*}[(p_d(\omega)^{1-\sigma})^2] = \sum_o \left(\frac{w_o(1+t_{od})}{A_{od}} \right)^{1-\sigma} \int_{z_{od}^*}^\infty z^{2(\sigma-1)} \theta^* z^{-\theta^*-1} dz.$$

The integral is only finite if $2(\sigma - 1) - \theta^* < 0$, which again gives $\theta^* > 2(\sigma - 1)$.

B Proofs

Throughout this Appendix we bound the gains from trade of a single country, which we label j . We write $\mathcal{A}(F)$ for the set of productivity shifters $\{A\}$ that rationalize the data under the distribution F (unique up to the normalization $P_d = 1$), and $\mathcal{V}$ for the distributions admitting a strictly positive density with a finite $(\sigma - 1)$ -th moment. We denote $\tau_{od} \equiv 1 + t_{od}$ and $V_{\tau\tau} \equiv \text{Var}(\Delta_o \Delta_d \ln \tau)$. For any bilateral variable M write

$$C_M \equiv \text{Cov}(\Delta_o \Delta_d M, \Delta_o \Delta_d \ln \tau),$$

B.1 Proof of Proposition 1(a)

Proposition 1(a) (Bounds for competitive model). *Suppose that the baseline trade data satisfy, for country j of interest, the condition:*

$$\lambda_{jj}/\lambda_{jd} \geq \lambda_{oj}/\lambda_{od} \quad \forall o, d \neq j \quad (\text{C1})$$

Consider all possible productivity distributions F . Then the smallest and largest possible values for j 's gains from trade in the competitive model, as defined in program (22), are given by

$$\underline{GT}_j = 0 \quad \text{and} \quad \overline{GT}_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (41)$$

Proof. Normalise $w_j = w'_j = 1$ and hold the deficit fixed as a share of income, so that by (12) and (2),

$$GT_j = 1 - \left(\frac{\gamma'_j}{\gamma_j} \right)^{1/(\sigma-1)}, \quad (42)$$

where γ_j and γ'_j are the factual and autarky values of the price aggregate (2) for j . Because GT_j is strictly decreasing in the autarky-to-trade ratio γ'_j/γ_j , it suffices to prove that in each model γ'_j/γ_j ranges over $(\lambda_{jj}, 1)$, with both endpoints sharp; the bounds on GT_j then follow from (42).

Setup. For a good ω and structural matrix A define the competitiveness of origin o in market d ,

$$\Phi_{od}(\omega) \equiv MC_{od}(\omega)^{1-\sigma} = \kappa_{od}^{1-\sigma} z_o(\omega)^{\sigma-1}, \quad \kappa_{od} \equiv \frac{w_o \tau_{od}}{A_{od}}.$$

By (4) the buyer in d purchases from $\arg \max_o \Phi_{od}$, so,

$$\gamma_d = \mathbb{E}_F[\max_o \Phi_{od}], \quad \lambda_{od} = \frac{\mathbb{E}_F[\Phi_{od} \mathbb{1}\{o \text{ wins } d\}]}{\gamma_d},$$

where “ o wins d ” is the event $\{\Phi_{od} \geq \Phi_{cd} \forall c\}$ (ties have F -measure zero and are ignored). In autarky j produces every good itself, so $\gamma'_j = \mathbb{E}_F[\Phi_{jj}]$.

Step 1 (universal bounds). Fix any $F \in \mathcal{V}$ with $A = \mathcal{A}(F)$. Pointwise, $\max_o \Phi_{oj} \geq \Phi_{jj} \geq 0$. The event $\{j \text{ loses } j\} = \{\exists o \neq j : \Phi_{oj} > \Phi_{jj}\}$ has strictly positive F -measure: otherwise $\mathbb{E}_F[\Phi_{oj} \mathbb{1}\{o \text{ wins } j\}] = 0$ for every $o \neq j$, forcing $\sum_{o \neq j} \lambda_{oj} = 0$, contradicting $\lambda_{jj} < 1$. On that event $\max_o \Phi_{oj} > \Phi_{jj}$ and $\Phi_{jj} > 0$ ($z_j > 0$ a.s. with positive density), whence

$$\mathbb{E}_F[\max_o \Phi_{oj}] > \mathbb{E}_F[\Phi_{jj}] \implies \gamma'_j / \gamma_j < 1. \quad (43)$$

Moreover $\lambda_{jj} \gamma_j = \mathbb{E}_F[\Phi_{jj} \mathbb{1}\{j \text{ wins } j\}] < \mathbb{E}_F[\Phi_{jj}] = \gamma'_j$, where the inequality is strict because

$\mathbb{E}_F[\Phi_{jj} \mathbb{1}\{j \text{ loses } j\}] > 0$ by the same argument; hence $\gamma'_j / \gamma_j > \lambda_{jj}$. Combining, $\gamma'_j / \gamma_j \in (\lambda_{jj}, 1)$, i.e.

$$0 < GT_j < 1 - \lambda_{jj}^{1/(\sigma-1)} \quad \text{for every } F \in \mathcal{V}. \quad (44)$$

Step 2 (thresholds and the losing boundary). Hold the productivity shifters fixed at the Frechet baseline $A^* = \mathcal{A}(F^*)$; by construction these satisfy the orthogonality restriction (20). Given competitors' draws $z_{-j} = (z_o)_{o \neq j}$, origin j wins market d iff $\Phi_{jd} \geq \Phi_{od}$ for all $o \neq j$, i.e. iff

$$z_j^{\sigma-1} \geq T_d(z_{-j}) \equiv \max_{o \neq j} \left(\frac{\kappa_{od}}{\kappa_{jd}} \right)^{1-\sigma} z_o^{\sigma-1},$$

for all κ evaluated at A^* . Define the losing boundary $B(z_{-j}) \equiv (\min_d T_d(z_{-j}))^{1/(\sigma-1)}$; that is, for $z_j < B$ origin j wins no market, and for $z_j \geq B$ it wins at least one.

Step 3 (Condition C1 pins the boundary at the home market). Under F^* , we have: $\lambda_{od} / \lambda_{jd} = (\kappa_{od} / \kappa_{jd})^{-\theta^*}$ for all o, d . Condition C1 is $\lambda_{jj} \lambda_{od} \geq \lambda_{oj} \lambda_{jd}$, or $\lambda_{od} / \lambda_{jd} \geq \lambda_{oj} / \lambda_{jj}$, which is equivalent to $(\kappa_{od} / \kappa_{jd})^{1-\sigma} \geq (\kappa_{oj} / \kappa_{jj})^{1-\sigma}$. Multiplying by $z_o^{\sigma-1} > 0$ and taking $\max_{o \neq j}$ we have,

$$T_j(z_{-j}) \leq T_d(z_{-j}) \quad \text{for all } d \neq j, z_{-j}, \quad \text{hence } B(z_{-j})^{\sigma-1} = T_j(z_{-j}). \quad (45)$$

In particular j wins its own market iff $z_j \geq B(z_{-j})$.

Step 4 (invariance of the data under reallocation below B). Let F have the same z_{-j} -marginal as F^* and a conditional law of z_j given z_{-j} that agrees with F^* on $[B(z_{-j}), \infty)$; the total conditional mass on $(0, B)$ is then automatically that of F^* . We claim every share is unchanged,

so $\mathcal{A}(F) = A^*$. Condition on z_{-j} and abbreviate $B = B(z_{-j})$. Since $\max_{o \neq j} \Phi_{od} = \kappa_{jd}^{1-\sigma} T_d$, we have $\max_c \Phi_{cd} = \kappa_{jd}^{1-\sigma} \max\{z_j^{\sigma-1}, T_d\}$, and $T_d \geq T_j = B^{\sigma-1}$ by (45); thus

$$\mathbb{E}_F[\max_c \Phi_{cd} \mid z_{-j}] = \kappa_{jd}^{1-\sigma} \left(T_d \mathbb{P}(z_j < T_d^{1/(\sigma-1)} \mid z_{-j}) + \mathbb{E}[z_j^{\sigma-1} \mathbb{1}\{z_j \geq T_d^{1/(\sigma-1)}\} \mid z_{-j}] \right).$$

Because $T_d^{1/(\sigma-1)} \geq B$, both the probability $\mathbb{P}(z_j < T_d^{1/(\sigma-1)}) = \mathbb{P}(z_j < B) + \mathbb{P}(B \leq z_j < T_d^{1/(\sigma-1)})$ and the truncated expectation over $[T_d^{1/(\sigma-1)}, \infty) \subseteq [B, \infty)$ depend on the conditional law only through its total mass on $(0, B)$ and its restriction to $[B, \infty)$, both preserved; so $\gamma_d = \mathbb{E}_F[\max_c \Phi_{cd}]$ is unchanged. For the numerator with $o = j$, $\mathbb{E}_F[\Phi_{jd} \mathbb{1}\{j \text{ wins } d\} \mid z_{-j}] = \kappa_{jd}^{1-\sigma} \mathbb{E}[z_j^{\sigma-1} \mathbb{1}\{z_j \geq T_d^{1/(\sigma-1)}\} \mid z_{-j}]$ is unchanged for the same reason. For $o \neq j$, let $\mathcal{Z}_o(d) = \{z_{-j} : \Phi_{od} \geq \Phi_{cd} \forall c \neq j\}$, which is a function of z_{-j} only. On $\mathcal{Z}_o(d)$, $T_d = (\kappa_{od}/\kappa_{jd})^{1-\sigma} z_o^{\sigma-1}$ and o wins d iff additionally $z_j^{\sigma-1} \leq T_d$, so

$$\mathbb{E}_F[\Phi_{od} \mathbb{1}\{o \text{ wins } d\}] = \kappa_{od}^{1-\sigma} \mathbb{E}_F[z_o^{\sigma-1} \mathbb{1}_{\mathcal{Z}_o(d)} \mathbb{P}(z_j \leq T_d^{1/(\sigma-1)} \mid z_{-j})],$$

again unchanged since $T_d^{1/(\sigma-1)} \geq B$. Hence all λ_{od} are unchanged and $\mathcal{A}(F) = A^*$. This means that F matches all data and satisfies (20).

Step 5 (the $[0, B(z_{-j})]$ contribution to GT_j). By Step 3, j wins j iff $z_j \geq B$, so $\lambda_{jj}\gamma_j = \kappa_{jj}^{1-\sigma} \mathbb{E}_F[z_j^{\sigma-1} \mathbb{1}\{z_j \geq B\}]$ and therefore

$$\gamma'_j = \mathbb{E}_F[\Phi_{jj}] = \lambda_{jj}\gamma_j + S(F), \quad S(F) \equiv \kappa_{jj}^{1-\sigma} \mathbb{E}_F[z_j^{\sigma-1} \mathbb{1}\{z_j < B\}] \geq 0, \quad (46)$$

$$\frac{\gamma'_j}{\gamma_j} = \lambda_{jj} + \frac{S(F)}{\gamma_j}, \quad \frac{S(F)}{\gamma_j} \in [0, 1 - \lambda_{jj}]. \quad (47)$$

Decomposing $\gamma_j = \mathbb{E}_F[\max_o \Phi_{oj}] = \mathbb{E}_{F^*}[\max_o \Phi_{oj}]$ over $\{z_j \geq B\}$ (winner j , contributing $\lambda_{jj}\gamma_j$) and $\{z_j < B\}$ (where, by (45), $\max_o \Phi_{oj} = \kappa_{jj}^{1-\sigma} T_j = \kappa_{jj}^{1-\sigma} B^{\sigma-1}$) gives the companion identity

$$\kappa_{jj}^{1-\sigma} \mathbb{E}_{F^*}[B^{\sigma-1} \mathbb{1}\{z_j < B\}] = (1 - \lambda_{jj})\gamma_j. \quad (48)$$

Step 6 (upper bound is sharp). For $\alpha \in (0, 1)$ let F_α reallocate the below- B mass toward 0:

$$f_\alpha(z \mid z_{-j}) = \begin{cases} (1 - \alpha) p(z_{-j}) B^{-(1-\alpha)} z^{-\alpha}, & 0 < z < B, \\ f^*(z \mid z_{-j}), & z \geq B, \end{cases} \quad p(z_{-j}) \equiv \mathbb{P}^{F^*}(z_j < B \mid z_{-j}).$$

This is a density: $\int_0^B (1 - \alpha) p B^{-(1-\alpha)} z^{-\alpha} dz = p B^{-(1-\alpha)} B^{1-\alpha} = p$ (finite as $\alpha < 1$), plus mass $1 - p$ on $[B, \infty)$; it is strictly positive and agrees with f^* on $[B, \infty)$ with below- B

mass p , so Step 4 gives $\mathcal{A}(F_\alpha) = A^*$. Its below- B $(\sigma - 1)$ -th moment is $\int_0^B z^{\sigma-1} f_\alpha dz = (1 - \alpha) p B^{-(1-\alpha) \frac{B^{\sigma-\alpha}}{\sigma-\alpha}} = \frac{1-\alpha}{\sigma-\alpha} p B^{\sigma-1}$ (finite as $\sigma - \alpha > 0$), so $F_\alpha \in \mathcal{V}$ and, by (48), $S(F_\alpha) = \frac{1-\alpha}{\sigma-\alpha} (1 - \lambda_{jj}) \gamma_j$. Hence by (47)

$$\frac{\gamma'_j}{\gamma_j} = \lambda_{jj} + \frac{1-\alpha}{\sigma-\alpha} (1 - \lambda_{jj}),$$

which, since $\frac{d}{d\alpha} \frac{1-\alpha}{\sigma-\alpha} = \frac{1-\sigma}{(\sigma-\alpha)^2} < 0$, decreases to λ_{jj} as $\alpha \uparrow 1$. By (42), $GT_j(F_\alpha) \uparrow 1 - \lambda_{jj}^{1/(\sigma-1)}$. With the strict ceiling (44), $\sup GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$, attained in the limit by F^+ (all below- B mass at 0), the case where j cannot reproduce its imports.

Step 7 (lower bound is sharp). For $n \in \mathbb{N}$ let F_n concentrate the below- B mass just under B : $f_n(z | z_{-j}) = n p(z_{-j}) B^{-n} z^{n-1}$ on $(0, B)$ and f^* on $[B, \infty)$; this is a density with below- B mass $\int_0^B n p B^{-n} z^{n-1} dz = p$, and below- B $(\sigma - 1)$ -th moment $\int_0^B z^{\sigma-1} n p B^{-n} z^{n-1} dz = \frac{n}{n+\sigma-1} p B^{\sigma-1}$, so $F_n \in \mathcal{V}$ and $\mathcal{A}(F_n) = A^*$. Then $S(F_n) = \frac{n}{n+\sigma-1} (1 - \lambda_{jj}) \gamma_j$ and $\gamma'_j / \gamma_j = \lambda_{jj} + \frac{n}{n+\sigma-1} (1 - \lambda_{jj}) \uparrow 1$ as $n \rightarrow \infty$, so $GT_j(F_n) \downarrow 0$. With (44), $\inf GT_j = 0$, attained in the limit by F^- (below- B mass at B^-), the case where j reproduces each imported good at the winning foreign price. Both F_α, F_n are absolutely continuous w.r.t. F^* ; this proves the bounds and their sharpness in the competitive model. $\square$

B.2 Proof of proposition 1(b)

Proposition 1(b) (Bounds for monopolistic competition model). *Suppose that the baseline tariff data satisfy the condition that,*

$$\exists i_1 \neq i_2, i_1 \neq j: \quad \Delta_o \Delta_d \ln(1 + t_{i_1 i_2}) > 0 \quad (\text{C2})$$

Consider all possible productivity distributions F . Then the smallest and largest possible values for j 's gains from trade in the monopolistic competition model, as defined in program (26), are given by

$$\underline{GT}_j = 0 \quad \text{and} \quad \overline{GT}_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (49)$$

Proof. Normalize $w_j = w'_j = 1$ and hold the deficit fixed as a share of income; then by (12)–(2) autarky expenditure equals factual expenditure, $E'_j = E_j$, and

$$GT_j = 1 - \left(\frac{\gamma'_j}{\gamma_j} \right)^{1/(\sigma-1)}. \quad (50)$$

It suffices to show that γ'_j / γ_j ranges over $(\lambda_{jj}, 1)$, with both endpoints sharp. Throughout,

the counterfactual obeys the two restrictions of program (26): the mass of entrants is unchanged, $N_o = N'_o$, and every home firm produces in autarky, $(\hat{z}_{jj})' = 1$.

Setup. Write $Y \equiv z^{\sigma-1}$ and, for the law F_o of Y at origin o ,

$$\mathcal{E}_F(k) \equiv \mathbb{E}_F[Y \mathbb{1}\{Y > k\}], \quad \bar{E}_F \equiv \mathbb{E}_F[Y] = \mathcal{E}_F(1).$$

The home marginal F_j is free in $\mathcal{V}$; the foreign marginals are i.i.d. Pareto on $(1, \infty)$ with shape $s \equiv \theta^*/(\sigma - 1) > 1$ (without loss for GT_j , which depends on the foreign blocks only through aggregates).

Step 1 (welfare under $N_o = N'_o$). A home firm with draw z charges $p_{jj} = \frac{\sigma}{\sigma-1} w_j / (A_{jj} z)$ (since there are no tariffs domestically), so $p_{jj}^{1-\sigma} = C Y$ with $C \equiv (\frac{\sigma}{\sigma-1} w_j / A_{jj})^{1-\sigma} > 0$, and it serves the home market iff $Y > k$, where $k \equiv \hat{z}_{jj}^{\sigma-1}$ is the factual home cutoff. By (15) the home contribution to the factual price aggregate is $N_j C \mathcal{E}_{F_j}(k)$, and being the share λ_{jj} of γ_j ,

$$\lambda_{jj} \gamma_j = N_j C \mathcal{E}_{F_j}(k). \quad (51)$$

In autarky j consumes only home varieties; with the entrant mass held at $N'_j = N_j$, all producing ($k' = 1$), and $w'_j = w_j$,

$$\gamma'_j = N_j C \mathcal{E}_{F_j}(1) = N_j C \bar{E}_{F_j}. \quad (52)$$

Dividing (52) by (51) (the common factor $N_j C$ cancels, as does E_j via $E'_j = E_j$),

$$\frac{\gamma'_j}{\gamma_j} = \frac{\lambda_{jj} \bar{E}_{F_j}}{\mathcal{E}_{F_j}(k)}. \quad (53)$$

Step 2 (universal bounds). Two inequalities pin down the range of (53). First, $\mathcal{E}_{F_j}(k) \leq \mathcal{E}_{F_j}(1) = \bar{E}_{F_j}$ for $k \geq 1$, so

$$\frac{\gamma'_j}{\gamma_j} \geq \lambda_{jj} \implies GT_j \leq 1 - \lambda_{jj}^{1/(\sigma-1)}. \quad (54)$$

Second, $(\hat{z}_{jj})' = 1$ requires the least-productive home firm ($Y = 1$) to exactly break even in autarky, $\frac{1}{\sigma} X'_{jj}(1) = w_j f_{jj}$. Using $X'_{jj}(Y) = E_j (\gamma'_j)^{-1} p_{jj}^{1-\sigma} = E_j C Y / \gamma'_j$ with (52), $X'_{jj}(1) = E_j / (N_j \bar{E}_{F_j})$; while the factual zero-cutoff-profit condition $\frac{1}{\sigma} X_{jj}(k) = w_j f_{jj}$ with (51) gives

$f_{jj} = k \lambda_{jj} E_j / (\sigma N_j \mathcal{E}_{F_j}(k))$. The break-even equality is therefore $\mathcal{E}_{F_j}(k) = k \lambda_{jj} \bar{E}_{F_j}$, so by (53)

$$\frac{\gamma'_j}{\gamma_j} = \frac{1}{k} \leq 1 \quad \implies \quad GT_j \geq 0. \quad (55)$$

Step 3 (sharpness). Take the home marginal Pareto with shape $m > 1$: $\mathcal{E}_{F_j}(k) = \frac{m}{m-1} k^{1-m}$, $\bar{E}_{F_j} = \frac{m}{m-1}$, so $\bar{E}_{F_j} / \mathcal{E}_{F_j}(k) = k^{m-1}$ and (53) reads $\gamma'_j / \gamma_j = \lambda_{jj} k^{m-1}$, while the break-even bound (55) becomes $k \leq \lambda_{jj}^{-1/m}$.

Upper bound: let $m \downarrow 1$ at cutoff $k = \lambda_{jj}^{-1/m}$; then $\gamma'_j / \gamma_j \downarrow \lambda_{jj}$ and $GT_j \downarrow 1 - \lambda_{jj}^{1/(\sigma-1)}$, the attaining F^+ .

Lower bound: let $m \rightarrow \infty$ at the break-even cutoff $k = \lambda_{jj}^{-1/m}$, so the home marginal concentrates at the floor; then $\gamma'_j / \gamma_j = \lambda_{jj}^{1/m} \uparrow 1$ and $GT_j = 1 - \lambda_{jj}^{1/(m(\sigma-1))} \downarrow 0$, the attaining F^- .

Step 4 (feasibility of trade flow and orthogonality constraints). It remains to exhibit, for each F_j above, a full array (A, f, f_E, N) that reproduces every bilateral flow and satisfies both orthogonality restrictions (20) and (21). Normalize $\gamma_d = 1$. A firm from o has sales $E_d p_{od}^{1-\sigma} = E_d Y / B_{od}$, $B_{od} \equiv (\frac{\sigma}{\sigma-1} w_o \tau_{od} / A_{od})^{\sigma-1}$ and operates iff $Y > k_{od} = B_{od} \sigma w_o f_{od} / E_d$; the trade share and fixed cost invert to

$$B_{od} = \frac{N_o \mathcal{E}_{F_o}(k_{od})}{\lambda_{od}}, \quad f_{od} = \frac{k_{od} \lambda_{od} E_d}{\sigma w_o N_o \mathcal{E}_{F_o}(k_{od})}. \quad (56)$$

These invert cell by cell: for any cutoff $k_{od} \geq 1$ the first relation in (56) fixes B_{od} and, equivalently, A_{od} . The second fixes f_{od} ; hence $(A_{od}, f_{od}) \mapsto (\lambda_{od}, k_{od})$ is a bijection at each cell, so the cutoff matrix $\{k_{od}\}$ is free. The origin scalars N_o (and f_{oE}) are fixed by the origins' free-entry conditions and enter only as origin-only terms. Write $\mu_{od} \equiv \Delta_o \Delta_d \ln k_{od}$, and set $\ln k_{jd} = \ln k$ constant in d , so that the home row of $\ln \mathcal{E}_{F_j}(k_{jd})$ is origin-only and drops under $\Delta_o \Delta_d$; on foreign rows $\ln \mathcal{E}_{F_o}(k_{od}) = \text{const}_o + (1-s) \ln k_{od}$. Applying $\Delta_o \Delta_d$ to the logarithms of (56) then yields

$$\Delta_o \Delta_d \ln A_{od} = \frac{1}{\sigma-1} \Delta_o \Delta_d \ln \lambda_{od} + \Delta_o \Delta_d \ln \tau_{od} + \frac{s-1}{\sigma-1} \mu_{od}, \quad (57)$$

$$\Delta_o \Delta_d \ln f_{od} = s \mu_{od} + \Delta_o \Delta_d \ln \lambda_{od}. \quad (58)$$

By the orthogonality moment (19), $C_\lambda = -\theta^* V_{\tau\tau} = -s(\sigma-1) V_{\tau\tau}$, so calculating the covariance of (57)–(58) and $\Delta_o \Delta_d \ln \tau$ yields (since $s > 1$) that both moments vanish iff the single equation

$$\text{Cov}(\mu, \Delta_o \Delta_d \ln \tau) = (\sigma-1) V_{\tau\tau} \quad (59)$$

holds. Set each foreign row constant in d , $\ln k_{od} = \alpha_o$ for $o \neq j$, at arbitrary interior values, free entry is then absorbed by the free origin scalars f_{Eo} , and place a single adjustment $\nu > 0$ at the pivot cell (i_1, i_2) of Condition **C2**, which lies on a foreign row since $i_1 \neq j$: $\nu \equiv (\sigma - 1)V_{\tau\tau}/C_{E^{i_1 i_2}}$, where $E^{i_1 i_2}$ is the unit matrix at the pivot cell and $C_{E^{i_1 i_2}}$ is a positive multiple of $\Delta_o \Delta_d \ln \tau_{i_1 i_2} > 0$. Off the pivot, every entry of $\ln k$ is origin-only and is removed by $\Delta_o \Delta_d$, so $\text{Cov}(\mu, \Delta_o \Delta_d \ln \tau) = \nu C_{E^{i_1 i_2}} = (\sigma - 1)V_{\tau\tau}$, which is (59). Every cutoff is interior, and the export-selection ranking $\hat{z}_{od} \geq \hat{z}_{oo}$ holds within every row, with equality off the pivot and a strict inequality at the pivot since $\nu > 0$. Thus for every $F_j \in \mathcal{V}$ the array (A, f, f_E, N) reproduces all bilateral flows, satisfies both orthogonality restrictions, and leaves all home firms producing in autarky, so it is feasible for program (26). With Steps 2–3 this proves $\sup_F GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$ and $\inf_F GT_j = 0$. $\square$

B.3 Proof of Proposition 2

Proposition 2 (Bounds for competitive model with independence). *Suppose that the baseline tariff and trade data satisfy the following condition: for Ξ a square matrix composed of elements Ξ_{od} that are zero apart from the j th row with elements given by $\frac{1}{1-\sigma} \ln \frac{\lambda_{jd}}{1-\lambda_{jd}}$, we have*

$$\text{Var}(\Delta_o \Delta_d \ln(1 + t_{od})) > \text{Cov}(\Delta_o \Delta_d \Xi_{od}, \Delta_o \Delta_d \ln(1 + t_{od})) \quad (\text{C3})$$

Consider all possible productivity distributions F under which countries draw their productivities independently. Then the largest possible value for country j 's gains from trade in the competitive model, as defined in program (22), is given by

$$\overline{GT}_j^{\text{indep.}} = \overline{GT}_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (60)$$

Further, this value is approached arbitrarily closely by a sequence of distributions with Frechet marginals.

Proof. From the proof of Proposition 1(a), with the numeraire $w_j = w'_j = 1$,

$$GT_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}} (1 - L)^{-\frac{1}{\sigma-1}}, \quad L \equiv \frac{\mathbb{E}_F[z_j(\omega)^{\sigma-1} \mathbb{1}\{j \text{ loses } j\}]}{\mathbb{E}_F[z_j(\omega)^{\sigma-1}]}, \quad (\text{I1})$$

where “ j loses j ” is the event $\{z_j(\omega)^{\sigma-1} < \widetilde{M}\}$ with $\widetilde{M} \equiv \kappa_{jj}^{\sigma-1} \max_{o \neq j} \kappa_{oj}^{1-\sigma} z_o(\omega)^{\sigma-1}$ (in the notation of Appendix B.1, $\widetilde{M} = T_j$ and $L = S(F)/\gamma'_j$). Since $L \in [0, 1 - \lambda_{jj}]$, equation (12) gives the universal ceiling $GT_j \leq 1 - \lambda_{jj}^{1/(\sigma-1)}$, approached as $L \downarrow 0$, for every F . It remains to exhibit independent-Frechet, data-consistent models with $L \rightarrow 0$ and $C_{\ln A} = 0$.

Construction. For each $n \geq 1$ let every country draw independently, with j drawing from $\text{Frechet}(\alpha_n)$, $\alpha_n \equiv (\sigma - 1) + \frac{1}{n}$, and each competitor $o \neq j$ drawing from a common $\text{Frechet}(\gamma)$, $\gamma > \sigma - 1$ fixed below. Since $\alpha_n > \sigma - 1$, $\mathbb{E}[z_j^{\sigma-1}] < \infty$ (so $F_n \in \mathcal{V}$); the scales, equivalently the shifters $A^{(n)}$ through $\kappa_{od} = w_o \tau_{od} / A_{od}$, are pinned down in Step 3.

Steps 1 (the gravity moment by an intermediate-value argument). Let $\kappa^{(n)}(\alpha_n, \gamma)$ be the effective-cost matrix matching all shares at shapes (α_n, γ) (Step 2 shows it exists and is unique), and put

$$G_n(\gamma) \equiv V_{\tau\tau} - C_{\ln \kappa^{(n)}(\alpha_n, \gamma)}. \quad (I2)$$

As $\ln \kappa_{od} = \ln w_o + \ln \tau_{od} - \ln A_{od}$ and $\Delta_o \Delta_d$ removes the origin-only $\ln w_o$, we have $\Delta_o \Delta_d \ln A_{od} = \Delta_o \Delta_d \ln \tau_{od} - \Delta_o \Delta_d \ln \kappa_{od}$, so $C_{\ln A} = V_{\tau\tau} - C_{\ln \kappa} = G_n$; thus the moment (20) is equivalent to $G_n(\gamma) = 0$.

Lower endpoint $G_n(\alpha_n) < 0$. At $\gamma = \alpha_n$ all shapes coincide, giving the single-elasticity **Eaton and Kortum (2002)** gravity model $\lambda_{od} = \kappa_{od}^{-\alpha_n} / \sum_{o'} \kappa_{o'd}^{-\alpha_n}$. Inverting, $\ln \kappa_{od} = -\frac{1}{\alpha_n} \ln \lambda_{od} + h_d$ with h_d destination-only, so $\Delta_o \Delta_d \ln \kappa_{od} = -\frac{1}{\alpha_n} \Delta_o \Delta_d \ln \lambda_{od}$ and $C_{\ln \kappa} = -\frac{1}{\alpha_n} C_{\ln \lambda} = \frac{\theta^*}{\alpha_n} V_{\tau\tau}$. Hence $G_n(\alpha_n) = V_{\tau\tau} (1 - \theta^* / \alpha_n) < 0$ for all n large enough that $\alpha_n < \theta^*$ (possible since $\theta^* > \sigma - 1$ and $\alpha_n \downarrow \sigma - 1$).

Upper endpoint $\lim_{\gamma \rightarrow \infty} G_n(\gamma) > 0$. For fixed γ the **Eaton and Kortum (2002)** closed form inverts the competitor entries to $\ln \kappa_{od}(\gamma) = \ln \Psi_d(\gamma) - \frac{1}{\gamma} \ln p_{od}$, $p_{od} \equiv \lambda_{od} / (1 - \lambda_{jd})$, $\Psi_d \equiv (\sum_{o \neq j} \kappa_{od}^{-\gamma})^{-1/\gamma}$, and the home entries to $\ln \kappa_{jd}(\gamma) = \ln \Psi_d(\gamma) - \frac{1}{\alpha_n} \ln s_d(\gamma)$, with $s_d(\gamma) \equiv (\Psi_d / \kappa_{jd})^{\alpha_n}$ pinned down by the home value share through a continuous, strictly increasing map $\lambda_{jd} = H_n(s_d; \gamma)$; explicitly, $H_n(s_d; \gamma) \equiv \mathbb{E}[Z \mathbb{1}\{Z > \tilde{M}\}] / \mathbb{E}[\max(Z, \tilde{M})]$ with $Z \sim \text{Frechet}(\theta_n, s_d^{1/\theta_n})$, $\tilde{M} \sim \text{Frechet}(\gamma/(\sigma - 1), 1)$, and $\theta_n \equiv \alpha_n / (\sigma - 1)$. The destination-only level $\ln \Psi_d$ is removed by $\Delta_o \Delta_d$. As $\gamma \rightarrow \infty$ the competitor terms $-\frac{1}{\gamma} \ln p_{od} \rightarrow 0$ while the home row tends to $-\frac{1}{\alpha_n} \ln s_d^\infty$, so $\ln \kappa_{od}^{(n)} \rightarrow \ln \Psi_d + \Xi_{od}^{(n)}$ pointwise, with $\Xi_{od}^{(n)} \equiv -\frac{1}{\alpha_n} \ln s_d^\infty \mathbb{1}\{o = j\}$ and $s_d^\infty \equiv \lim_{\gamma \rightarrow \infty} s_d(\gamma) = (H_n^\infty)^{-1}(\lambda_{jd})$. As $\gamma \rightarrow \infty$ the competitors' maximum welfare-weighted productivity degenerates at its scale while j remains $\text{Frechet}(\alpha_n)$; $H_n(\cdot; \gamma)$ converges to a strictly increasing H_n^∞ , and the inverse of a continuous strictly monotone function depends continuously on it, so $s_d(\gamma) \rightarrow s_d^\infty$. Explicitly, $H_n^\infty(s_d) = s_d^{1/\theta_n} \gamma_{\text{low}}(1 - \frac{1}{\theta_n}, s_d) / (s_d^{1/\theta_n} \gamma_{\text{low}}(1 - \frac{1}{\theta_n}, s_d) + e^{-s_d})$, with $\gamma_{\text{low}}(a, x) \equiv \int_0^x v^{a-1} e^{-v} dv$ the lower incomplete gamma (which is finite since $\theta_n > 1$). Since $\Delta_o \Delta_d$ is a fixed linear (and hence continuous) operator removing the destination-only $\ln \Psi_d$,

$$\lim_{\gamma \rightarrow \infty} \Delta_o \Delta_d \ln \kappa_{od}^{(n)} = \Delta_o \Delta_d \Xi^{(n)}. \quad (61)$$

The leading-order expansion as $\alpha_n \downarrow \sigma - 1$ gives $\frac{1}{\alpha_n} \ln s_d^\infty = \frac{1}{\sigma - 1} \ln \frac{\lambda_{jd}}{1 - \lambda_{jd}}$ up to a destination-

independent constant; on the home row that constant is a pure origin effect, simplified by $\Delta_o \Delta_d$, so $\Delta_o \Delta_d \Xi^{(n)} \rightarrow \Delta_o \Delta_d \Xi$ as $n \rightarrow \infty$. Therefore $\lim_{\gamma \rightarrow \infty} G_n(\gamma) = V_{\tau\tau} - C_{\Xi^{(n)}} \rightarrow V_{\tau\tau} - C_{\Xi} > 0$ by **C3**. Thus for all large n , G_n is continuous on $[\alpha_n, \infty)$ with $G_n(\alpha_n) < 0 < \lim_{\gamma \rightarrow \infty} G_n(\gamma)$, and the intermediate value theorem yields a finite root $\gamma_n^* \in (\alpha_n, \infty)$ with $G_n(\gamma_n^*) = 0$, i.e. the moment (20) is satisfied.

Step 2 (exact matching of all trade flows). Fix (α_n, γ_n^*) . Among competitors $c, k \neq j$ the shares satisfy $\lambda_{cd}/\lambda_{kd} = (\kappa_{cd}/\kappa_{kd})^{-\gamma_n^*}$, while the home share λ_{jd} is smooth and strictly decreasing in κ_{jd} from 1 to 0. These supply $D(D-1)$ independent equations in the $D(D-1)$ unknown ratios of $\kappa^{(n)}$ (with the normalisation $\kappa_{jj} = 1$); by the inverse function theorem there is a unique $\kappa^{(n)}$, hence a unique $A_{od}^{(n)} = \kappa_{od}^{(n)}/(w_o \tau_{od})$, reproducing every bilateral flow. The moment (20) holds because $G_n(\gamma_n^*) = 0$, and the $(\sigma-1)$ -th moment is finite since $\alpha_n > \sigma-1$. Thus F_n is an admissible, independent, Fréchet-marginal model satisfying all constraints of program (22).

Step 3 (the welfare limit $L_n \rightarrow 0$). Let $Y \equiv z_j(\omega)^{\sigma-1} \sim \text{Frechet}(\theta_n)$, $\theta_n \equiv \alpha_n/(\sigma-1) = 1 + \frac{1}{n(\sigma-1)} \downarrow 1$, and let $\widetilde{M}_n$ be the competitor maximum in j 's market as in (I1); Y and $\widetilde{M}_n$ are independent. The two share identities at $d = j$,

$$\lambda_{jj} = \frac{\mathbb{E}[Y \mathbb{1}\{Y > \widetilde{M}_n\}]}{\mathbb{E}[\max(Y, \widetilde{M}_n)]}, \quad 1 - \lambda_{jj} = \frac{\mathbb{E}[\widetilde{M}_n \mathbb{1}\{\widetilde{M}_n \geq Y\}]}{\mathbb{E}[\max(Y, \widetilde{M}_n)]},$$

give $\mathbb{E}[\max(Y, \widetilde{M}_n)] = \mathbb{E}[Y](1 - L_n)/\lambda_{jj}$ and hence, using $\max(Y, \widetilde{M}_n) \geq \widetilde{M}_n$, $\mathbb{E}[\widetilde{M}_n]/\mathbb{E}[Y] \leq 1/\lambda_{jj}$. With $\widehat{Y}_n \equiv Y/\mathbb{E}[Y]$ ($\mathbb{E}[\widehat{Y}_n] = 1$) and $\widehat{M}_n \equiv \widetilde{M}_n/\mathbb{E}[Y]$ ($\mathbb{E}[\widehat{M}_n] \leq 1/\lambda_{jj}$), the loss fraction is $L_n = \mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq \widehat{M}_n\}]$. By independence and Fubini, for any fixed $K > 0$,

$$\begin{aligned} L_n &= \int_0^\infty y \mathbb{P}(\widehat{M}_n \geq y) f_{\widehat{Y}_n}(y) dy \leq \mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq K\}] + \int_K^\infty y \cdot \frac{\mathbb{E}[\widehat{M}_n]}{y} f_{\widehat{Y}_n}(y) dy \\ &\leq \mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq K\}] + \frac{\mathbb{P}(\widehat{Y}_n > K)}{\lambda_{jj}}, \end{aligned}$$

applying Markov's inequality $\mathbb{P}(\widehat{M}_n \geq y) \leq \mathbb{E}[\widehat{M}_n]/y \leq 1/(\lambda_{jj}y)$ on $[K, \infty)$, where the factor y cancels against $1/y$. Fix K and let $n \rightarrow \infty$. Since $\mathbb{E}[Y] = \Gamma(1 - 1/\theta_n) \sim 1/(\theta_n - 1) = n(\sigma-1) \rightarrow \infty$,

$$\mathbb{P}(\widehat{Y}_n > K) = 1 - \exp(-(K \mathbb{E}[Y])^{-\theta_n}) \rightarrow 0,$$

and the truncated mean $\mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq K\}] = \Gamma(1 - \frac{1}{\theta_n}, (K \mathbb{E}[Y])^{-\theta_n})/\Gamma(1 - \frac{1}{\theta_n}) \rightarrow 0$, because the bulk of the $\text{Frechet}(\theta_n)$ mass concentrates near 0 as $\theta_n \downarrow 1$ while its diverging mean is

carried by a vanishing-probability upper tail. More precisely, the ratio is the regularized upper incomplete gamma $Q(s_n, x_n) = \Gamma(s_n, x_n)/\Gamma(s_n)$ with $s_n \equiv 1 - \frac{1}{\theta_n} \in (0, 1)$ and $x_n \equiv (\text{KE}[Y])^{-\theta_n}$: as $n \rightarrow \infty$, $s_n \downarrow 0$ and $x_n \downarrow 0$; the expansion $\gamma_{\text{low}}(s, x) = \frac{x^s}{s}(1 + O(x))$ gives $Q(s, x) = 1 - \frac{x^s}{\Gamma(s+1)}(1 + O(x))$, and since $\theta_n s_n = \theta_n - 1$, $x_n^{s_n} = \exp(-(\theta_n - 1) \ln(\text{KE}[Y]))$ with $\mathbb{E}[Y] \sim (\theta_n - 1)^{-1}$, so $(\theta_n - 1) \ln(\text{KE}[Y]) \rightarrow 0$; thus $x_n^{s_n} \rightarrow 1$, $\Gamma(s_n + 1) \rightarrow 1$, and $Q \rightarrow 0$. Both terms vanish for every fixed K , so $L_n \rightarrow 0$. By (I1), $GT_j^{(n)} = 1 - \lambda_{jj}^{1/(\sigma-1)}(1 - L_n)^{-1/(\sigma-1)} \rightarrow 1 - \lambda_{jj}^{1/(\sigma-1)}$; with the universal ceiling this gives $\sup_F GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$, approached but not attained. $\square$

B.4 Proof of Proposition 3

Condition 1. Fix the country of interest j . There is a partition $\mathcal{P} = \{S_1, \dots, S_K\}$ of the competitor set $O \setminus \{j\}$ such that

$$\sum_{k: C_{\psi S_k} < 0} \text{Cov}(\Delta_o \Delta_d \psi^{S_k}, \Delta_o \Delta_d \ln(t_{od} + 1)) < \text{Var}(\Delta_o \Delta_d \ln(t_{od} + 1)) (\sigma - 1 - \theta^*).$$

With

$$\psi_{od}^S \equiv \mathbb{1}\{o \in S\} \ln \frac{\lambda_{od}}{\Lambda_{S,d}}, \quad \Lambda_{S,d} \equiv \sum_{o' \in S} \lambda_{o'd}$$

($\psi^S \equiv 0$ when $|S| = 1$).

Proposition 3 (Bounds for competitive model with common marginal distributions). Suppose that the baseline tariff and trade data satisfy the data condition 1. Consider all possible productivity distributions F under which countries draw their productivities from a common marginal distribution but with potential dependence across countries. Then the largest possible value for country j 's gains from trade in the competitive model, as defined in program (22), is given by

$$\overline{GT}_j^{\text{CM}} = \overline{GT}_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (62)$$

Further, this bound is approached arbitrarily closely by a distribution with Frechet(1, θ) marginals.

Proof. **Step 1 (the nested construction and the covariance identity).** Partition $O \setminus \{j\} = S_1 \cup \dots \cup S_K$ and fix $\rho = (\rho_1, \dots, \rho_K) \in [0, 1]^K$. Let (X_o) follow the two-level nested-logistic (GEV) max-stable law

$$\mathbb{P}(X \leq x) = \exp\left(-x_j^{-\beta} - \sum_{k=1}^K \left(\sum_{o \in S_k} x_o^{-\beta/(1-\rho_k)}\right)^{1-\rho_k}\right),$$

under which j and the blocks $S_1, \dots, S_K$ are independent at the top level and each block S_k is a logistic nest of dependence ρ_k . Every marginal is *Frechet*(1, β), with $\beta \equiv \theta/(\sigma - 1)$, so $z_o = X_o^{1/(\sigma-1)} \sim \text{Frechet}(1, \theta)$; at $\rho = 0$ this is the i.i.d. law. Denote it $F_{\theta, \rho}^{\mathcal{P}}$. Writing $b_{od} \equiv \kappa_{od}^{1-\sigma}$, each offer is $\Phi_{od} = b_{od} X_o$. Substituting $x_o = \phi/b_{od}$ into the displayed CDF (the event $\{\Phi_{od} \leq \phi \forall o\}$) collapses every argument to ϕ :

$$\mathbb{P}(\Phi_d^{\max} \leq \phi) = \exp(-\Theta_d \phi^{-\beta}), \quad \Theta_d = b_{jd}^\beta + \sum_k T_{S_k}(d), \quad T_{S_k}(d) = D_{kd}^{1-\rho_k}, \quad D_{kd} \equiv \sum_{o \in S_k} b_{od}^{\beta/(1-\rho_k)},$$

where the block term used $(\sum_{o \in S_k} (\phi/b_{od})^{-\beta/(1-\rho_k)})^{1-\rho_k} = \phi^{-\beta} D_{kd}^{1-\rho_k}$. Thus the global best offer is $\Phi_d^{\max} \sim \text{Frechet}(\Theta_d, \beta)$, the home offer is $\text{Frechet}(b_{jd}^\beta, \beta)$, and the block maximum $\max_{o \in S_k} \Phi_{od} \sim \text{Frechet}(T_{S_k}(d), \beta)$, all independent. For a GEV max-stable law the identity of the argmax is independent of the value of the maximum, so the expenditure share equals the choice probability; the nested-logit factorisation (nest choice $\times$ within-nest choice) gives

$$\lambda_{jd} = \frac{b_{jd}^\beta}{\Theta_d}, \quad \lambda_{od} = \underbrace{\frac{T_{S_k}(d)}{\Theta_d}}_{= \Lambda_{S_k, d}} \underbrace{\frac{b_{od}^{\beta/(1-\rho_k)}}{D_{kd}}}_{= \lambda_{od} / \Lambda_{S_k, d}} \quad (o \in S_k), \quad \Lambda_{S_k, d} \equiv \sum_{o \in S_k} \lambda_{od}.$$

Taking logs of the two competitor factors and using $b_{od}^\beta = \kappa_{od}^{(1-\sigma)\beta} = \kappa_{od}^{-\theta}$ (so $\frac{\beta}{1-\rho_k} \ln b_{od} = -\frac{\theta}{1-\rho_k} \ln \kappa_{od}$),

$$\psi_{od}^{S_k} = \ln \frac{\lambda_{od}}{\Lambda_{S_k, d}} = -\frac{\theta}{1-\rho_k} \ln \kappa_{od} - \ln D_{kd}, \quad \ln \Lambda_{S_k, d} = (1-\rho_k) \ln D_{kd} - \ln \Theta_d.$$

Since $\ln \lambda_{od} = \ln \Lambda_{S_k, d} + \psi_{od}^{S_k}$, the combination that eliminates $\ln D_{kd}$ is

$$\ln \lambda_{od} - \rho_k \psi_{od}^{S_k} = \ln \Lambda_{S_k, d} + (1-\rho_k) \psi_{od}^{S_k} = -\theta \ln \kappa_{od} - \ln \Theta_d.$$

The home relation $\lambda_{jd} = \kappa_{jd}^{-\theta} / \Theta_d$ inverts the shares to

$$\ln \kappa_{od} = -\frac{1}{\theta} (\ln \lambda_{od} - \rho_k \psi_{od}^{S_k}) - \frac{1}{\theta} \ln \Theta_d \quad (o \in S_k), \quad \ln \kappa_{jd} = -\frac{1}{\theta} \ln \lambda_{jd} - \frac{1}{\theta} \ln \Theta_d.$$

Since $\ln A_{od} = -\ln \kappa_{od} + \ln w_o + \ln \tau_{od}$ we have,

$$\Delta_o \Delta_d \ln A_{od} = \frac{1}{\theta} \Delta_o \Delta_d (\ln \lambda_{od} - \Psi_{od}^{\mathcal{P}, \rho}) + \Delta_o \Delta_d \ln \tau_{od}, \quad \Psi^{\mathcal{P}, \rho} \equiv \sum_k \rho_k \psi^{S_k}.$$

Pairing with $\Delta_o \Delta_d \ln \tau$ and using $C_{\ln \lambda} = -\theta^* V_{\tau\tau}$ gives the covariance identity, affine in ρ ,

$$K(\mathcal{A}(F_{\theta,\rho}^{\mathcal{P}})) = \frac{1}{\theta} \left(C_{\ln \lambda} - \sum_k \rho_k C_{\psi^{S_k}} \right) + V_{\tau\tau}.$$

Step 2 (Verifying the gravity covariance condition down to $\sigma - 1$). $K(\mathcal{A}(F_{\theta,\rho}^{\mathcal{P}})) = 0$ is the single linear equation

$$\sum_k \rho_k C_{\psi^{S_k}} = V_{\tau\tau} (\theta - \theta^*). \quad (63)$$

Set $\rho_k = 0$ on every block with $C_{\psi^{S_k}} \geq 0$ and let $\rho_k \in [0, 1)$ on the negative blocks; the left side then sweeps the interval $(\sum_{k: C_{\psi^{S_k}} < 0} C_{\psi^{S_k}}, 0]$, where (approached as those $\rho_k \uparrow 1$). For $\theta \in (\sigma - 1, \theta^*]$ the right side lies in $[V_{\tau\tau}(\sigma - 1 - \theta^*), 0]$; under condition 1, $\sum_{k: C_{\psi^{S_k}} < 0} C_{\psi^{S_k}} < V_{\tau\tau}(\sigma - 1 - \theta^*)$, so the whole target interval is reached and (63) has a solution $\rho(\theta) \in [0, 1)^K$ for every $\theta \in (\sigma - 1, \theta^*]$ (e.g. uniform $\rho_k = V_{\tau\tau}(\theta - \theta^*) / \sum_{k: C_{\psi^{S_k}} < 0} C_{\psi^{S_k}}$ on the negative blocks). Each $F_{\theta,\rho(\theta)}^{\mathcal{P}}$ has common *Frechet*(1, θ) marginals, finite $(\sigma - 1)$ -th moments (since $\theta > \sigma - 1$), matches all bilateral trade flows by the inversion, and satisfies $K = 0$; it therefore obeys all constraints of program (22) with arbitrary (GEV) cross-country dependence.

Step 3 (welfare is the gravity value at θ , free of dependence). From the proof of Proposition 1(a),

$$GT_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}} \left(\frac{\mathbb{E}_F[z_j(\omega)^{\sigma-1}]}{\mathbb{E}_F[z_j(\omega)^{\sigma-1} \mathbb{1}\{j \text{ wins } j\}]} \right)^{\frac{1}{\sigma-1}}.$$

In every distribution constructed above, z_j is drawn independently of $(z_o)_{o \neq j}$ and all marginals are *Frechet*(1, θ). The home offer $b_{jj} X_j \sim \text{Frechet}(b_{jj}^\beta, \beta)$ and the competitor maximum $\max_{o \neq j} b_{oj} X_o$ are independent, the latter *Frechet*(T, β) for some scale T . By the [Eaton and Kortum \(2002\)](#) value identity for two independent common-shape Frechet offers,

$$\mathbb{E}_F[z_j^{\sigma-1} \mathbb{1}\{j \text{ wins } j\}] / \mathbb{E}_F[z_j^{\sigma-1}] = \lambda_{jj}^{1-1/\beta} = \lambda_{jj}^{1-(\sigma-1)/\theta}, \text{ whence}$$

$$GT_j(F_{\theta,\rho(\theta)}^{\mathcal{P}}) = 1 - \lambda_{jj}^{1/\theta}.$$

GT_j is free of the dependence, which enters only through the scale T .

Step 4 (the limit). $GT_j(F_{\theta,\rho(\theta)}^{\mathcal{P}}) = 1 - \lambda_{jj}^{1/\theta} \uparrow 1 - \lambda_{jj}^{1/(\sigma-1)}$ as $\theta \downarrow \sigma - 1$. Combined with the universal ceiling $GT_j \leq 1 - \lambda_{jj}^{1/(\sigma-1)}$ (valid for every F), this gives $\sup_F GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$. $\square$

B.5 Proof of Proposition 4

The proof exhibits i.i.d. models in which, market by market, the home country undercuts every rival on all goods it can meaningfully produce, so that the welfare-loss fraction L of (I1) is bounded by the (vanishing) probability weight of the marginal's irregular components; the gravity moment is restored exactly through the near-zero mass ϖ by an intermediate-value argument. Existence and uniqueness of the rationalizing array $\mathcal{A}(F)$ are maintained; what the argument additionally requires of the inversion is that it vary continuously in ϖ and that it converge, as the marginal degenerates, to an explicit two-block profile. Both follow from the variational structure established in Lemma 1; the limiting profile is characterized in Lemma 2, using Rado's theorem.

Classical results. We use three classical results, stated here for completeness and invoked by name below.

Rado's theorem. Let $r \in \mathbb{R}^m$ and let $\Pi(r)$ denote the convex hull of the $m!$ vectors obtained by permuting the coordinates of r . Then $x \in \Pi(r)$ if and only if

$$\sum_{k=1}^m x_k = \sum_{k=1}^m r_k \quad \text{and} \quad \sum_{k \leq k'} x_{(k)} \leq \sum_{k \leq k'} r_{(k)} \quad (k' = 1, \dots, m-1),$$

where $x_{(1)} \geq \dots \geq x_{(m)}$ is the decreasing rearrangement, that is, if and only if x is *majorized* by r (Rado, 1952; for a textbook treatment, Marshall, Olkin, and Arnold, 2011, Ch. 2). Only the “if” direction, ajorization implies membership in $\Pi(r)$, is used below.

Rearrangement inequality. Let $a_1 \geq \dots \geq a_m$ and $b_1 \geq \dots \geq b_m$. For every permutation σ of $\{1, \dots, m\}$,

$$\sum_k a_k b_{m+1-k} \leq \sum_k a_k b_{\sigma(k)} \leq \sum_k a_k b_k,$$

and the upper bound is attained exactly at similarly ordered pairings: if σ pairs some $a_i > a_j$ with $b_{\sigma(i)} < b_{\sigma(j)}$, swapping those two assignments strictly increases the sum (Hardy, Littlewood, and Pólya, 1952, Theorem 368).

Berge's maximum theorem. Let T be a metric space, K a nonempty compact subset of a metric space, and $f : T \times K \rightarrow \mathbb{R}$ jointly continuous. Then $v(t) = \min_{u \in K} f(t, u)$ is continuous, and $t \mapsto \arg \min_{u \in K} f(t, u)$ is upper hemicontinuous with nonempty compact values; in particular, if the minimizer is unique for every t , it is a continuous function of t (Berge, 1963; see also Aliprantis and Border, 2006, Theorem 17.31). We apply it with a constant constraint set K ; equicoercivity is what confines all minimizers to such a set.

Notation. As in the proof of Proposition 3, write $b_{od} \equiv \kappa_{od}^{1-\sigma}$ and $X_o \equiv z_o^{\sigma-1}$, so that $\Phi_{od} = b_{od}X_o$; under i.i.d. draws the common law of X_o is denoted G . Shares are column-separable: $(\lambda_{od})_o$ depends on b only through the column $(b_{od})_o$, up to a destination scale that $\Delta_o\Delta_d$ annihilates. Let $m \equiv D - 1$ and, for $\omega \in (0, 1)$,

$$\chi(\omega) \equiv \ln \frac{\omega(1-\omega^m)}{1-\omega}, \quad \Delta_d(\omega) \equiv \ln \frac{\lambda_{dd}}{1-\lambda_{dd}} + \chi(\omega), \quad R_k(\omega) \equiv \frac{1-\omega^k}{1-\omega^m} \quad (k = 1, \dots, m-1),$$

and for each destination d define the normalized import shares $q_{od} \equiv \lambda_{od}/(1-\lambda_{dd})$, $o \neq d$, with decreasing rearrangement $q_{(1)d} \geq \dots \geq q_{(m)d}$. Finally, let E^{dd} denote the $D \times D$ matrix whose only nonzero entry is a one in cell (d, d) , let $g_d \equiv C_{E^{dd}}$, and let Ξ^ω be the diagonal matrix with entries $\Xi_{dd}^\omega \equiv \frac{1}{1-\omega} \Delta_d(\omega)$.

Condition 2 (Two-block data condition). *There is an interval $[\omega_1, \omega_2] \subset (0, 1)$ such that:*

$$\Delta_d(\omega_1) > 0 \quad \text{for every } d; \quad (\text{C4a})$$

$$\sum_{k \leq k'} q_{(k)d} < R_{k'}(\omega_2) \quad \text{for every } d \text{ and every } k' = 1, \dots, m-1; \quad (\text{C4b})$$

$$(C_{\Xi^{\omega_1}} - V_{\tau\tau})(C_{\Xi^{\omega_2}} - V_{\tau\tau}) < 0. \quad (\text{C4c})$$

Because C_M is linear in M and Ξ^ω is affine in $\chi(\omega)$, condition (C4c) is a one-dimensional root location: $C_{\Xi^\omega} - V_{\tau\tau} = -\frac{1}{\sigma-1}(P + \chi(\omega)Q) - V_{\tau\tau}$ with the data scalars $P \equiv \sum_d \ln(\lambda_{dd}/(1-\lambda_{dd}))g_d$ and $Q \equiv \sum_d g_d$. Since χ increases strictly from $-\infty$ to $\ln m$ on $(0, 1)$ (Lemma 2(iii)), if $Q < 0$ the root is unique and available in closed form, $\omega_\infty = \chi^{-1}(-[(\sigma-1)V_{\tau\tau} + P]/Q)$, and (C4c) holds for any interval containing it. Note also the symmetry with condition (C3): the matrix Ξ there carries the transform $\frac{1}{1-\sigma} \ln(\lambda_{jd}/(1-\lambda_{jd}))$ on the focal row; Ξ^ω carries the same transform of the home shares on the diagonal, shifted by the common scalar $\chi(\omega)$.

Proposition 4 (Bounds for the competitive model with i.i.d. productivities). *Suppose the baseline tariff and trade data satisfy Condition 2. Across all $F \in \mathcal{V}$ under which countries draw productivities independently from a common marginal distribution, the upper bound on GT_j in program (22) is*

$$\overline{GT}_j^{\text{i.i.d.}} = 1 - \lambda_{jj}^{1/(\sigma-1)}.$$

The bound is approached arbitrarily closely along a sequence of i.i.d. distributions whose common marginal has a strictly positive density on $(0, \infty)$, and it is not attained.

The potential and the normalization. Fix a destination d (suppressed where possible), write $u_o \equiv \ln b_{od}$, $b = e^u$, and

$$\varphi(u) \equiv \ln \Psi(u), \quad \Psi(u) \equiv \mathbb{E} \left[\max_o b_o X_o \right], \quad M_{-o} \equiv \max_{c \neq o} b_c X_c \text{ (independent of } X_o \text{)}.$$

Costs in a column are identified only up to a destination scale: Ψ is homogeneous of degree one in b , so $\varphi(u + c\mathbf{1}) = \varphi(u) + c$ for every $c \in \mathbb{R}$, and share columns sum to one, so the data cannot distinguish u from $u + c\mathbf{1}$. We fix the scale once and for all, normalizing every cost column to the hyperplane

$$V_0 \equiv \left\{ u \in \mathbb{R}^D : \sum_o u_o = 0 \right\}.$$

The choice is innocuous downstream: everything the proof extracts from the inversion, coordinate differences $u_d - u_o$, and the double-differenced objects entering C_M , is invariant to it. The potential φ itself remains defined on all of $\mathbb{R}^D$, where its coordinate partials are the expenditure shares.

Lemma 1 (Exact derivative of the potential; the inversion as a convex program). *Let the common law G of X_o have finite mean and $G \neq \delta_0$.*

(i) *For every u and o , the one-sided partial derivatives of φ exist and equal*

$$\frac{\partial^+ \varphi}{\partial u_o} = \frac{\mathbb{E}[b_o X_o \mathbf{1}\{b_o X_o \geq M_{-o}\}]}{\Psi}, \quad \frac{\partial^- \varphi}{\partial u_o} = \frac{\mathbb{E}[b_o X_o \mathbf{1}\{b_o X_o > M_{-o}\}]}{\Psi},$$

and their difference equals

$$\frac{\partial^+ \varphi}{\partial u_o} - \frac{\partial^- \varphi}{\partial u_o} = \frac{b_o \mathbb{E}[X_o \mathbf{1}\{b_o X_o = M_{-o} > 0\}]}{\Psi} \geq 0 :$$

the expected value carried by the tie event.

(ii) *If the law of X_c / X_o given $X_o, X_c > 0$ is atomless, then value-carrying ties are null at every u , the gap in (i) vanishes, φ is C^1 , and*

$$\frac{\partial \varphi}{\partial u_o} = \frac{\mathbb{E}[b_o X_o \mathbf{1}\{o \text{ wins}\}]}{\Psi} = \lambda_{od}(u) \quad \text{exactly, with no boundary term,}$$

under any tie-breaking convention. Moreover $\sum_o \partial \varphi / \partial u_o = 1$, and if in addition G has a strictly positive density on $(0, \infty)$ then $\lambda_{od}(u) > 0$ for every o .

(iii) *φ is convex on $\mathbb{R}^D$ and strictly convex on V_0 , and the objective $\varphi(u) - \sum_o \lambda_{od} u_o$ is coercive on V_0 for any interior target column. Consequently the program $\min_{u \in V_0} \{\varphi(u) - \sum_o \lambda_{od} u_o\}$*

has a unique minimizer, and, under the hypothesis of (ii), $u \in V_0$ solves it if and only if the model reproduces the column exactly.

Proof. (i) For a fixed realization, $\psi(t) \equiv \max(b_o e^t X_o, M_{-o})$ is convex in t , so its difference quotients $[\psi(t) - \psi(0)]/t$ are nondecreasing in t and the one-sided limits exist pointwise; for $|t| \leq 1$ they are bounded by $e b_o X_o$, which is integrable. Pointwise,

$$\psi'(0^+) = b_o X_o \mathbb{1}\{b_o X_o \geq M_{-o}\}, \quad \psi'(0^-) = b_o X_o \mathbb{1}\{b_o X_o > M_{-o}\} :$$

on the tie, an increase in u_o puts o strictly ahead (slope $b_o X_o$), a decrease puts it strictly behind (slope 0). Monotone-dominated convergence passes the limits under $\mathbb{E}$, and the chain rule through $\ln$ gives the displays. Ties at zero ($b_o X_o = M_{-o} = 0$) carry zero value on both sides, whence the restriction > 0 in the gap.

It is worth recording why no Dirac term survives in the first derivative. Condition on $M_{-o} = \mu > 0$ and let G have density g . With $h(b) \equiv b \mathbb{E}[X \mathbb{1}\{X > \mu/b\}] + \mu \mathbb{P}(X \leq \mu/b)$, Leibniz differentiation produces two boundary terms, the mass $\frac{\mu}{b^2} g(\frac{\mu}{b})$ crossing the kink, valued at $b \cdot \frac{\mu}{b} = \mu$ on the winning branch and at μ on the losing branch:

$$h'(b) = \mathbb{E}[X \mathbb{1}\{bX > \mu\}] + \frac{\mu^2}{b^2} g(\frac{\mu}{b}) - \frac{\mu^2}{b^2} g(\frac{\mu}{b}),$$

which cancel exactly because $\max$ is value-matched at the kink; a surviving Dirac would require a jump in the objective, and there is none. The tie density reappears one derivative higher, differentiating the indicator concentrates on the tie set, so the Hessian of φ is a value-weighted density of ties, and, when value-carrying ties have positive probability, as the gap in (i): a genuine kink. The degenerate limit of Lemma 2 sits exactly on such a kink.

(ii) Under the atomless-ratio hypothesis, $\mathbb{P}(b_o X_o = b_c X_c > 0) = 0$ for $o \neq c$, hence $\mathbb{P}(b_o X_o = M_{-o} > 0) = 0$ at every u : the gap vanishes and the common value is $\mathbb{E}[b_o X_o \mathbb{1}\{o \text{ wins}\}]/\Psi$ under any tie convention (conventions differ on a value-null set). Continuity of the partials in u follows by dominated convergence, the winning indicator being a.s. continuous in u because value-ties are null at every u ; hence φ is C^1 . The unit sum is Euler's identity for the degree-one homogeneous $\Psi(b)$. Positivity under a strictly positive density: for each o , the event that X_o is large and every X_c , $c \neq o$, is small has positive probability for any b .

(iii) Convexity: pointwise $\max_o e^{(v_o + v'_o)/2} X_o \leq (\max_o e^{v_o} X_o)^{1/2} (\max_o e^{v'_o} X_o)^{1/2}$, and the Cauchy-Schwarz inequality gives $\varphi(\frac{v+v'}{2}) \leq \frac{1}{2}\varphi(v) + \frac{1}{2}\varphi(v')$. In the equality case $\max_o e^{v_o} X_o = c \max_o e^{v'_o} X_o$ a.s.; restricting to the positive-probability event on which a

given coordinate o attains both maxima yields $e^{v_o - v'_o} = c$ for every o , i.e. $v - v' \in \mathbb{R}\mathbf{1}$; for $v, v' \in V_0$ this means $v = v'$, so φ is strictly convex on V_0 . Coercivity on V_0 : $\Psi \geq e^{\max_o u_o} \mathbb{E}[X]$ (bound the max below by the coordinate attaining $\max_o u_o$), so

$$\varphi(u) - \sum_o \lambda_{od} u_o \geq \left(\min_o \lambda_{od} \right) \left(\max_o u_o - \min_o u_o \right) + \ln \mathbb{E}[X].$$

A minimizer therefore exists; at the minimum the first-order condition reads $\nabla \varphi(u) - \lambda_{\cdot d} \perp V_0$, and since both vectors have unit coordinate sums their difference also lies in V_0 , hence vanishes: $\nabla \varphi(u) = \lambda_{\cdot d}$, exact share matching. Uniqueness is strict convexity on V_0 ; conversely any share-matching $u \in V_0$ is a stationary point of a convex objective, hence the minimizer. $\square$

The degenerate limit. Let $\tilde{r}_k \equiv (1 - \omega)\omega^{k-1}$, $k = 1, \dots, D$, and define the degenerate potential and its branches

$$\varphi_\omega^\circ(u) \equiv \ln \left(\sum_{k=1}^D \tilde{r}_k e^{u^{(k)}} \right), \quad g_\pi(u) \equiv \ln \left(\sum_o \tilde{r}_{\pi(o)} e^{u_o} \right),$$

with $u_{(1)} \geq \dots \geq u_{(D)}$ the decreasing rearrangement and π ranging over the cost orderings (bijections) of $\{1, \dots, D\}$. Part (i) of the next lemma identifies φ_ω° as the potential of the limit law $\omega \delta_0 + (1 - \omega) \delta_1$: the economy in which a country is able to produce a given good with probability $1 - \omega$, all able producers are equally productive, and each good is supplied by its cheapest able producer.

Lemma 2 (Two-block limit profile). *Fix $\omega \in (0, 1)$ and an interior target column $(\lambda_{od})_o$.*

(i) $\varphi_\omega^\circ(u) = \ln \mathbb{E}[\max_o e^{u_o} S_o]$ for S_o i.i.d. Bernoulli($1 - \omega$), and $\varphi_\omega^\circ = \max_\pi g_\pi$. Each branch g_π is smooth and convex, with Hessian

$$\nabla^2 g_\pi(u) = \text{diag}(p(u)) - p(u) p(u)^\top, \quad p_o(u) = \frac{\tilde{r}_{\pi(o)} e^{u_o}}{\sum_c \tilde{r}_{\pi(c)} e^{u_c}},$$

and, for every $0 \neq h \in V_0$, $h^\top \nabla^2 g_\pi(u) h = \text{Var}_p(h) \geq (\min_o p_o(u)) \|h\|_2^2 > 0$. Both g_π and φ_ω° are strictly convex on V_0 , and for every interior target column the program $\min_{u \in V_0} \{\varphi_\omega^\circ(u) - \sum_o \lambda_{od} u_o\}$ has a unique minimizer.

(ii) If $\Delta_d(\omega) > 0$ and $\sum_{k \leq k'} q_{(k)d} \leq R_{k'}(\omega)$ for all $k' < m$, the minimizer is the normalized two-block column $\bar{u}^{(d)}(\omega) \equiv u^*(\omega) - \frac{\Delta_d(\omega)}{D} \mathbf{1} \in V_0$, where $u_d^*(\omega) = \Delta_d(\omega)$ and $u_o^*(\omega) = 0$ for $o \neq d$.

(iii) Δ_d is strictly increasing, with $\Delta'_d = \chi' \geq 1/\omega$; each $R_{k'}$ is strictly decreasing; $\chi(0^+) =$

$-\infty$ and $\chi(1^-) = \ln m$. If (C4a) holds at ω_1 and (C4b) holds at ω_2 , then the hypotheses of (ii) hold at every $\omega \in [\omega_1, \omega_2]$.

Proof. (i) Identities. Order the coordinates decreasingly (ties broken by index). The maximum $\max_o e^{u_o} S_o$ equals $e^{u^{(k)}}$ exactly when the origins carrying the $k-1$ largest coordinates are unable and the k -th is able, an event of probability $\omega^{k-1}(1-\omega) = \tilde{r}_k$ by independence, and equals 0 when all are unable; taking expectations gives the first identity. For the second, apply the rearrangement inequality (stated above) with $a = (e^{u^{(k)}})_k$ and $b = (\tilde{r}_k)_k$, both decreasing: $\sum_o \tilde{r}_{\pi(o)} e^{u_o} \leq \sum_k \tilde{r}_k e^{u^{(k)}}$ for every ordering π , with equality exactly at the similarly ordered ones, so $\max_\pi g_\pi = \varphi_\omega^\circ$.

Convexity. The Hessian formula is the log-sum-exp computation $\partial p_o / \partial u_c = \delta_{oc} p_o - p_o p_c$. For the variance bound, with $\bar{h} \equiv \frac{1}{D} \sum_o h_o$: $\text{Var}_p(h) = \sum_o p_o (h_o - \mathbb{E}_p h)^2 \geq (\min_o p_o) \sum_o (h_o - \mathbb{E}_p h)^2 \geq (\min_o p_o) \sum_o (h_o - \bar{h})^2$, the last step because the flat mean minimizes $a \mapsto \sum_o (h_o - a)^2$; for $h \in V_0$, $\bar{h} = 0$ and the right side is $(\min_o p_o) \|h\|_2^2$; positivity uses that every weight $\tilde{r}_k$ is strictly positive. A finite max of functions strictly convex on V_0 is strictly convex on V_0 (evaluate the active branch at the midpoint).

Existence and uniqueness. $\varphi_\omega^\circ(u) \geq \max_o u_o + \ln(1-\omega)$ (keep only the top term of the sum), so coercivity on V_0 follows as in Lemma 1(iii) with $\ln(1-\omega)$ in place of $\ln \mathbb{E}[X]$; a minimizer exists, and it is unique by strict convexity on V_0 .

(ii) Every object below is invariant to the common shift $u \mapsto u + c\mathbf{1}$, only coordinate differences enter, so we compute at the representative u^* rather than at $\bar{u}^{(d)}$. At u^* the home coordinate strictly exceeds the tied foreign coordinates; in the language of Lemma 1(i), home is off the tie set, while the m foreign coordinates sit on the kink. A branch evaluates $\sum_o \tilde{r}_{\pi(o)} e^{u_o^*} = \tilde{r}_{\pi(d)} (e^{\Delta_d} - 1) + \sum_k \tilde{r}_k$, strictly increasing in the weight $\tilde{r}_{\pi(d)}$ since $\Delta_d > 0$; as the weights are strictly decreasing, the active orderings are exactly those with $\pi(d) = 1$, home ranked cheapest, the foreign origins receiving $\{2, \dots, D\}$ in any order, with common value

$$Z \equiv e^{\Delta_d} (1 - \omega) + \omega (1 - \omega^m)$$

(geometric sum: $\sum_{k \geq 2} \tilde{r}_k = \omega(1 - \omega^m)$), and a branch with $\pi(d) = k \geq 2$ falls short by $(\tilde{r}_1 - \tilde{r}_k)(e^{\Delta_d} - 1) > 0$. For an active π ,

$$\nabla g_\pi(u^*) = \left(\frac{\tilde{r}_{\pi(o)} e^{u_o^*}}{Z} \right)_o : \\ \text{home entry } \frac{e^{\Delta_d}(1-\omega)}{Z}, \quad \text{foreign entries a permutation of } \frac{\omega(1-\omega^m)}{Z} (r_1, \dots, r_m),$$

where $r_k \equiv \tilde{r}_{k+1} / (\omega(1 - \omega^m))$ is decreasing with $\sum_k r_k = 1$ and $\sum_{k \leq k'} r_k = R_{k'}(\omega)$. By the

definition of Δ_d ,

$$e^{\Delta_d}(1-\omega) = \frac{\lambda_{dd}}{1-\lambda_{dd}} \omega(1-\omega^m) \implies Z = \frac{\omega(1-\omega^m)}{1-\lambda_{dd}}, \quad \frac{e^{\Delta_d}(1-\omega)}{Z} = \lambda_{dd}, \quad \frac{\omega(1-\omega^m)}{Z} = 1 - \lambda_{dd} :$$

every active gradient has home entry exactly λ_{dd} , and foreign block $(1 - \lambda_{dd})$ times a permutation of r . Since r is decreasing with $\sum_k r_k = 1 = \sum_k q_{(k)d}$, the assumed inequalities say exactly that the normalized import shares are majorized by r ; by Rado's theorem (stated above), the vector $(q_{od})_{o \neq d}$ lies in $\Pi(r)$, the convex hull of the permutations of r . Hence there are weights $\mu_\pi \geq 0$, $\sum_\pi \mu_\pi = 1$, on active orderings with $\sum_\pi \mu_\pi \nabla g_\pi(u^*) = (\lambda_{od})_o$. Each active branch satisfies $g_\pi \leq \varphi_\omega^\circ$ with equality at u^* , so for every v ,

$$\varphi_\omega^\circ(v) - \sum_o \lambda_{od} v_o \geq \sum_\pi \mu_\pi [g_\pi(u^*) + \nabla g_\pi(u^*) \cdot (v - u^*)] - \sum_o \lambda_{od} v_o = \varphi_\omega^\circ(u^*) - \sum_o \lambda_{od} u_o^*,$$

i.e. u^* is a global minimizer over $\mathbb{R}^D$; its V_0 representative $\bar{u}^{(d)}(\omega)$ attains the same value, hence is the unique minimizer of the normalized program, by (i).

(iii) $\chi'(\omega) = \frac{1}{\omega} + \frac{1}{1-\omega} - \frac{m\omega^{m-1}}{1-\omega^m} \geq \frac{1}{\omega}$ since $1 - \omega^m = (1 - \omega) \sum_{k < m} \omega^k \geq (1 - \omega)m\omega^{m-1}$; the limits follow from the geometric sum $\omega(1 - \omega^m)/(1 - \omega) = \omega \sum_{k < m} \omega^k \rightarrow m$ as $\omega \uparrow 1$. The derivative of $R_{k'}$ has the sign of $m\omega^{m-k'}(1 - \omega^{k'}) - k'(1 - \omega^m)$, which is strictly increasing in ω (its derivative is $m(m - k')\omega^{m-k'-1}(1 - \omega^{k'}) > 0$) and vanishes at $\omega = 1$, hence is negative on $(0, 1)$. Monotonicity extends (C4a) rightward from ω_1 and (C4b) leftward from ω_2 . $\square$

Proof of Proposition 4. By (I1) and (44), $GT_j < 1 - \lambda_{jj}^{1/(\sigma-1)}$ for every $F \in \mathcal{V}$, and $GT_j \rightarrow 1 - \lambda_{jj}^{1/(\sigma-1)}$ along any feasible sequence with $L \rightarrow 0$. It remains to exhibit i.i.d. members of $\mathcal{V}$, reproducing every bilateral flow and satisfying (20), along which $L \rightarrow 0$.

The family. Fix $\iota > 0$ (small, chosen below) and for $n \geq 1$ set $\zeta_n = \eta_n = 1/n$. Let the common law of $X_o \equiv z_o^{\sigma-1}$ be the three-part mixture

$$G_{\omega,n} \equiv \omega \text{Unif}(0, \zeta_n] + (1 - \omega - \eta_n) \text{Unif}[1, 1 + \iota] + \eta_n G^\circ, \quad \omega \in [\omega_1, \omega_2],$$

where G° is the law of $z^{\sigma-1}$ under $\text{Frechet}(1, \theta^*)$, with density strictly positive on $(0, \infty)$ and mean $\mu^\circ < \infty$ (finite because $\theta^* > \sigma - 1$). Draws are i.i.d. across countries, so $F_{\omega,n} \in \mathcal{V}$: the joint density is strictly positive on $(0, \infty)^D$ through the G° component, and the $(\sigma - 1)$ -th moment of z is the mean of X , which is finite. Call a draw *low*, *regular*, or *spread* according to its mixture component.

Exact matching and the gravity moment. The alive ratios of $G_{\omega,n}$ are atomless, so Lemma 1(ii)–(iii) applies: for every (ω, n) the convex programs (one per destination) deliver the nor-

malized effective-cost matrix $\kappa^{(n)}(\omega)$ reproducing every share, hence (with expenditures at their data values) every bilateral flow; the shifters are read off as $A_{od}^{(n)} = w_o \tau_{od} / \kappa_{od}^{(n)}$. As in the proof of Proposition 2, $\Delta_o \Delta_d \ln A = \Delta_o \Delta_d \ln \tau - \Delta_o \Delta_d \ln \kappa$, so the orthogonality moment (20) is equivalent to $K_n(\omega) \equiv C_{\ln \kappa^{(n)}(\omega)} - V_{\tau\tau} = 0$; the choice of normalization is irrelevant because $\Delta_o \Delta_d$ removes destination constants.

Uniform two-block limit. Let $u^{(d)}(\omega; n) \in V_0$ be the minimizing column and $\varphi_{\omega, n}$ the potential. Against the degenerate potential φ_ω° of Lemma 2: regular draws lie in $[1, 1 + \iota]$; low draws add at most $e^{\max_o u_o} \zeta_n$ to the maximum; spread draws at most $e^{\max_o u_o} \sum_o X_o \mathbb{1}\{o \text{ spread}\}$ in expectation; and carving the η_n weight out of the regular component rescales each $\tilde{r}_k$ by a factor in $[1 - \frac{\eta_n}{1-\omega_2}, (1 + \frac{\eta_n}{\omega_1})^D]$. With $\mathbb{E}[\max_o e^{u_o} S_o] \geq (1 - \omega) e^{\max_o u_o}$ in the degenerate model,

$$\sup_u |\varphi_{\omega, n}(u) - \varphi_\omega^\circ(u)| \leq \ln(1 + \iota) + \ln\left(1 + \frac{\zeta_n + D\eta_n u^\circ}{1 - \omega_2}\right) + D \ln\left(1 + \frac{\eta_n}{\omega_1}\right) + \ln \frac{1}{1 - \eta_n / (1 - \omega_2)} \equiv \varepsilon(\iota, n),$$

uniformly over $\omega \in [\omega_1, \omega_2]$, with $\varepsilon(\iota, n) \rightarrow \ln(1 + \iota)$ as $n \rightarrow \infty$. The objectives are equicoercive on V_0 ($\varphi_{\omega, n}(u) - \sum_o \lambda_{od} u_o \geq (\min_o \lambda_{od})(\max_o u_o - \min_o u_o) + \ln(1 - \omega_2 - \eta_n)$), so all minimizers, at every (ω, n) , and of the degenerate program, lie in one fixed compact set $K \subset V_0$. Write $\bar{u}^{(d)}(\omega)$ for the normalized two-block column of Lemma 2(ii). Two consequences.

First, at each fixed n the map $\omega \mapsto u^{(d)}(\omega; n)$ is continuous: $(\omega, u) \mapsto \varphi_{\omega, n}(u)$ is jointly continuous (dominated convergence in the mixture weights), minimizers are unique (Lemma 1(iii)), and Berge's maximum theorem (stated above, with the constant constraint set K) applies; hence $K_n(\cdot)$ is continuous on $[\omega_1, \omega_2]$.

Second, the finite- n minimizers are quantitatively close to the two-block profiles, by strict convexity of the limit rather than by a compactness argument, at fixed ι the objectives converge only to within $\ln(1 + \iota)$ of the degenerate ones. For $s > 0$ define the excess-growth modulus

$$m(s) \equiv \inf_{\omega \in [\omega_1, \omega_2]} \min_d \inf \left\{ [\varphi_\omega^\circ(v) - \sum_o \lambda_{od} v_o] - [\varphi_\omega^\circ(\bar{u}^{(d)}) - \sum_o \lambda_{od} \bar{u}_o^{(d)}] : v \in K, \|v - \bar{u}^{(d)}(\omega)\|_\infty = s \right\}.$$

Each inner infimum is over a compact sphere on which the excess is continuous and strictly positive (Lemma 2(i)–(ii)), it depends continuously on ω , and the outer infimum is over a compact interval: $m(s) > 0$. By convexity the excess at distance $\geq s$ from $\bar{u}^{(d)}$ is at least $m(s)$. Now if $\sup_u |\varphi_{\omega, n}(u) - \varphi_\omega^\circ(u)| \leq \varepsilon$, the minimizer $u^{(d)}(\omega; n)$ has degenerate excess at most 2ε (it beats $\bar{u}^{(d)}$ by ≤ 0 in the finite- n objective, and each evaluation moves

by at most ε), so

$$\|u^{(d)}(\omega; n) - \bar{u}^{(d)}(\omega)\|_\infty \leq s \quad \text{whenever} \quad 2\varepsilon(\iota, n) < m(s),$$

uniformly over $\omega \in [\omega_1, \omega_2]$ and d . Finally, since $\ln \kappa = \frac{1}{1-\sigma} \ln b$ column by column and $\Delta_o \Delta_d$ removes destination constants, the value of the gravity residual at the two-block profiles is $K_\infty(\omega) \equiv C_{\Xi\omega} - V_{\tau\tau}$ (their log-cost matrix is $\sum_d \Delta_d(\omega) E^{dd}$ up to destination constants), and, with C the norm of the linear functional $M \mapsto C_M$,

$$\sup_\omega |K_n(\omega) - K_\infty(\omega)| \leq \frac{C}{\sigma-1} \max_d \sup_\omega \|u^{(d)}(\omega; n) - \bar{u}^{(d)}(\omega)\|_\infty.$$

The orthogonality root. Set $3\delta \equiv \min_d \Delta_d(\omega_1) > 0$ by (C4a), and put

$$s^* \equiv \min \left\{ \frac{\delta}{2}, \frac{\sigma-1}{2C} \min_{i=1,2} |K_\infty(\omega_i)| \right\} > 0.$$

Choose ι small enough that both $\ln(1+\iota) < \delta$ and $4\ln(1+\iota) < m(s^*)$ —the order of choice matters, since $\varepsilon(\iota, n) \rightarrow \ln(1+\iota) > 0$ rather than to zero, and then n_0 such that $\varepsilon(\iota, n) \leq 2\ln(1+\iota)$ for all $n \geq n_0$. For $n \geq n_0$: every inverted column is within $s^* \leq \delta/2$ of its two-block profile, uniformly on $[\omega_1, \omega_2]$; consequently $\sup_\omega |K_n - K_\infty| \leq \frac{C}{\sigma-1} s^* \leq \frac{1}{2} \min_i |K_\infty(\omega_i)|$, so $K_n(\omega_1)$ and $K_n(\omega_2)$ are nonzero with the opposite signs given by (C4c). By continuity of K_n the intermediate value theorem yields $\omega^{(n)} \in (\omega_1, \omega_2)$ with $K_n(\omega^{(n)}) = 0$. The model $F_n \equiv F_{\omega^{(n)}, n}$ thus matches every bilateral flow exactly and satisfies (20) exactly: it is feasible for program (22).

The home cost gap. Coordinate differences are normalization-free, so by the uniform column bound ($s^* \leq \delta/2$, each of the two coordinates moving by at most $\delta/2$) and Lemma 2(iii), in every column d and for all $o \neq d$,

$$\ln b_{dd} - \ln b_{od} \geq \Delta_d(\omega^{(n)}) - \delta \geq \Delta_d(\omega_1) - \delta \geq 2\delta > \ln(1+\iota), \quad \text{i.e.} \quad b_{dd} > (1+\iota) b_{od}.$$

The welfare limit. In market j : if j 's draw is regular and every rival's draw is regular or low, j wins, a regular rival offers at most $b_{oj}(1+\iota) < b_{jj} \leq b_{jj} X_j$, a low rival at most $b_{oj} \zeta_n \leq b_{oj} < b_{jj}$. Hence $\{j \text{ loses } j\}$ requires j low or spread, or some rival spread, and

$$\mathbb{E}[X_j \mathbf{1}\{j \text{ loses } j\}] \leq \omega^{(n)} \zeta_n + \eta_n \mu^\circ + (D-1) \eta_n \mathbb{E}[X_j], \quad \mathbb{E}[X_j] \geq 1 - \omega_2 - \eta_n,$$

using independence for the rival-spread term, so

$$L_n \leq \frac{\omega_2 \zeta_n + \eta_n \mu^\circ}{1 - \omega_2 - \eta_n} + (D - 1)\eta_n \longrightarrow 0.$$

By (I1), $GT_j(F_n) \rightarrow 1 - \lambda_{jj}^{1/(\sigma-1)}$; with (44), the supremum over the i.i.d. subclass of $\mathcal{V}$ equals the ceiling, approached but not attained. $\square$